



[MDP211s]

Machine Elements Design

Smart Flow Meter

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Contribution sheet

Prototype Contribution sheet

Table 1 Prototype Contribution Table

Name	%
Moaz Gamal ALSayed	13.5
Abdullah Mahmoud Abdelkarim	14
Abdelkarim Wael Abdelkarim	14
Mohamed Islam Salah Aldin Bodour	13
Musa Gamal El-Shenawy	13.5
Mahmoud Salama Mohamed Abdo	12.5
Zeyad Samer Lotfy	13.5
Abdalrhman Medhat Maamoun	6

Report Contribution sheet

Table 2 Report Contribution sheet

Name	%
Moaz Gamal ALSayed	11
Abdullah Mahmoud Abdelkarim	10
Abdelkarim Wael Abdelkarim	13
Mohamed Islam Salah Aldin Bodour	15.5
Musa Gamal El-Shenawy	17
Mahmoud Salama Mohamed Abdo	10
Zeyad Samer Lotfy	12.5
Abdalrhman Medhat Maamoun	11

Abstract

This project presents the design and development of a smart meter for measuring the mass flow rate of air using three types of sensors.

The primary goal of this system is to accurately monitor and measure air flow in various engineering applications.

The smart meter employs a combination of advanced sensor technologies to capture data on air velocity and density, which is then used to calculate the mass flow rate. The system integrates these sensors with a microcontroller and software to process the measurements and provide real-time, digital readings. This design ensures high precision, reliability, and ease of use, making it suitable for a wide range of applications.

The smart meter features a user-friendly interface and is designed to provide instant digital readouts, making it accessible for both professional and general use.

The proposed solution also considers practical aspects such as cost-efficiency, ease of installation, and minimal maintenance, making it an ideal choice for both small-scale and large-scale applications. This report details the development process, technical design choices, sensor selection, and the performance evaluation of the system.

Introduction

The primary components of the flow meter include a pipe, two flanges for fixation, and three key sensors: a velocity sensor, a temperature sensor, and a pressure sensor.

By combining velocity, temperature, and pressure sensors, this smart flow meter delivers a reliable, cost-effective solution for accurate and efficient air flow measurement. This system is particularly useful in environments where precise airflow control is essential, such as in industrial processes, ventilation systems, and environmental monitoring.

The velocity sensor is strategically positioned to measure the rotational speed of the rotor, which is used to determine the velocity of the air passing through the pipe.

The temperature sensor measures the air temperature, and the pressure sensor provides data on the air's pressure. These three measurements—velocity, temperature, and pressure—are essential for calculating the mass flow rate of the air, which then outputs the mass flow rate in real-time through a digital display or communication interface.



Figure 1 The Designed Smart Flow Meter

Project Management

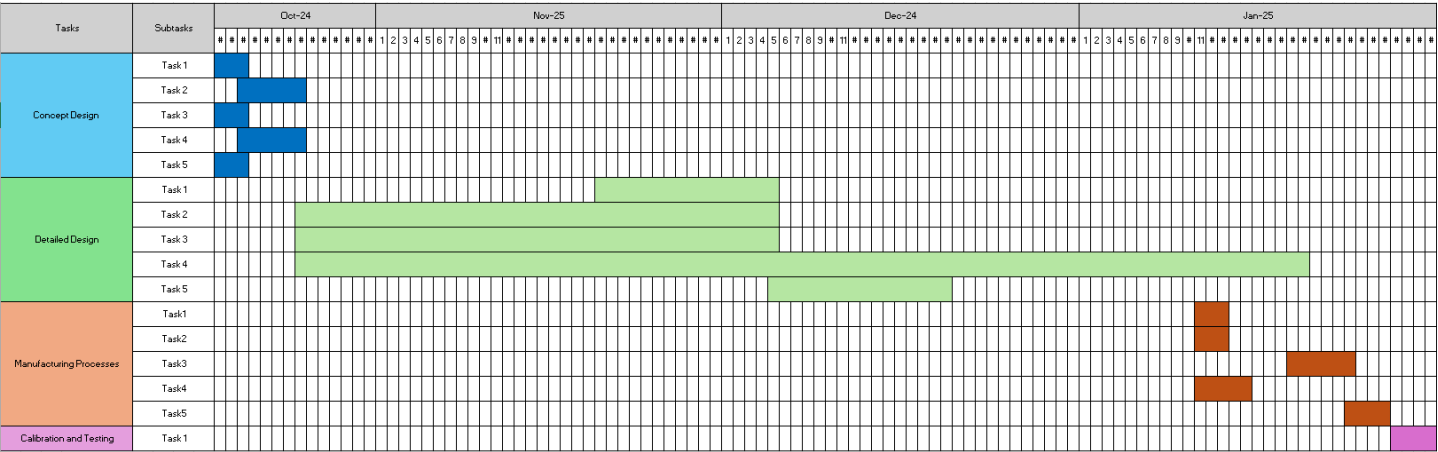


Figure 2 Gannt Chart

Table 3 Project Management

TASK	Discription	START	END
Concept Design			
Task 1	Rotor Fixation (CAP, Axial)	18/10/2024	20/10/2024
Task 2	Rotor Design For min Pressure Drop	20/10/2024	25/10/2024
Task 3	Temperature Sensor Selection	18/10/2024	20/10/2024
Task 4	Velocity Sensor Selection	20/10/2024	25/10/2024
Task 5	Pressure Sensor Selection	18/10/2024	20/10/2024
Detailed Design			
Task 1	CFD	20/11/2024	05/12/2024
Task 2	Final Rotor Design	25/10/2024	05/12/2024
Task 3	Sensors Fixation	25/10/2024	05/12/2024
Task 4	Electronics Component	25/10/2024	20/01/2025
Task 5	Housing+Hanger	05/12/2024	20/12/2025
Manufacturing Processes			
Task 1	Buying Pipe, Flanges And Sealing	11/01/2025	13/01/2025
Task 2	Buying St37 Rod for Hanger	11/01/2025	13/01/2025
Task 3	Manufacuring Rotor and Hanger and Welding	19/01/2025	24/01/2025
Task 4	Buying Velocity, Temperature and Pressure Sensors	11/01/2025	15/1/2025
Task 5	Housing Manufacuring and Assembly	24/01/2025	27/01/2025
Calibration and Testing		27/1/25	31/1/25

Drive Link:-

<https://docs.google.com/spreadsheets/d/1EZkzdOKJHjlnU4Nf968i0WbMV-YFepfL/edit?usp=sharing&ouid=115424722139153313570&rtpof=true&sd=true>

Good Points in Design:

Assembly and Disassembly for Easy Maintenance

- The smart flow meter is designed with a modular assembly, allowing for quick and effortless disassembly of components. This feature enhances maintenance efficiency by enabling easy access to internal parts for inspection, cleaning, and replacement. The streamlined design minimizes downtime, ensuring continuous and reliable operation.

Rotor Design

- The rotor is constructed using a **low-inertia material, specifically aluminum**, to optimize performance. This choice significantly reduces the starting torque, allowing for higher sensitivity and improved response to low flow rates. The lightweight nature of aluminum ensures minimal resistance to fluid movement, enhancing overall measurement accuracy.

Laser Welding for Precision and Low Fluid Turbulence

- The integration of **laser welding technology** ensures a seamless and highly precise assembly of components. This technique eliminates unnecessary gaps or imperfections in the structure, resulting in a **smooth internal flow path**. By minimizing turbulence within the fluid, laser welding enhances measurement accuracy and prolongs the operational lifespan of the flow meter. Additionally, the robust welds contribute to structural integrity, reducing the risk of leaks or mechanical failures over time.

Schematic For Smart air Flowmeter

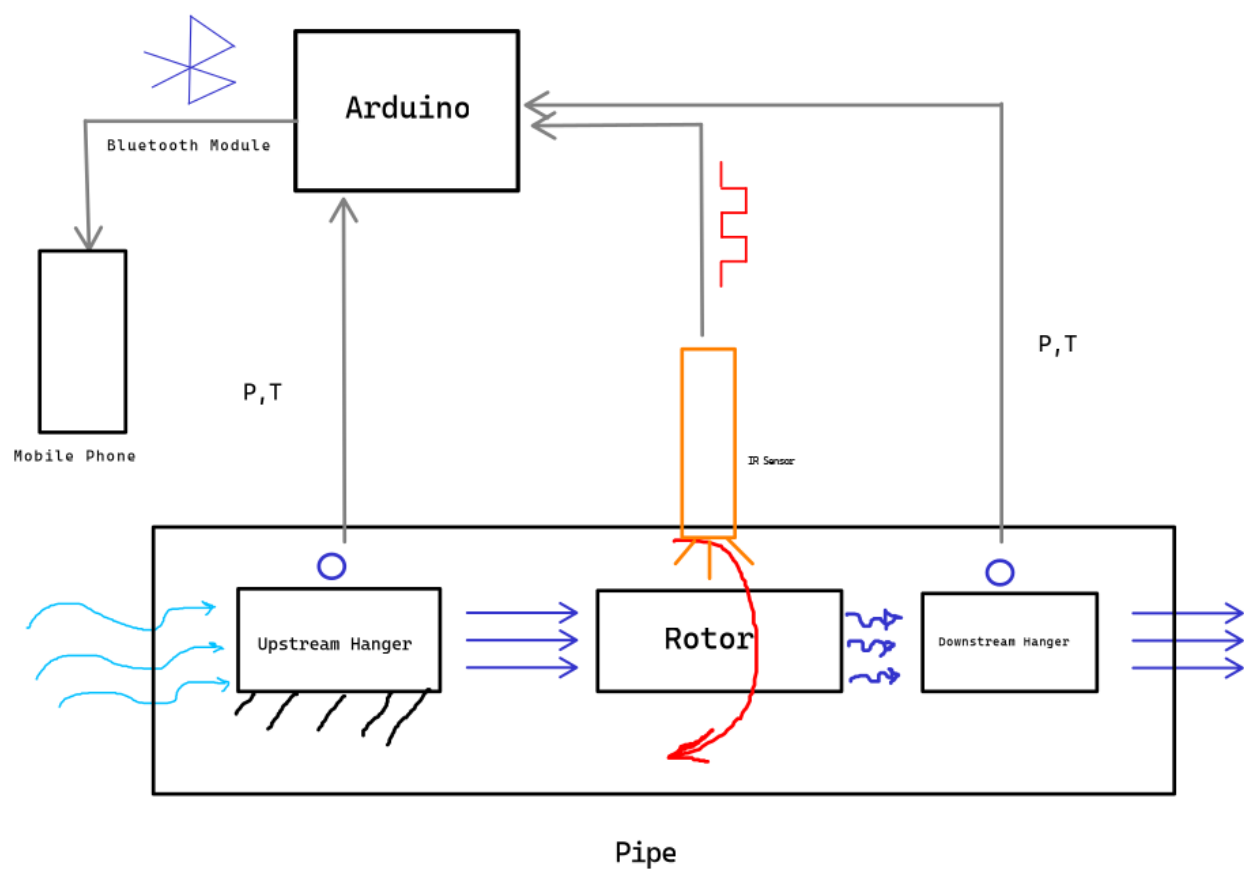


Figure 3 Schematic Diagram

Fluid and Turbomachinery

There are different approaches to calibrate the flow meter and derive a relation between the turbine rpm and the mass flow rate

1. Experimental
2. Numerical solution
3. Mathematical model

In this part we will discuss the numerical analysis in this analysis we will use ANSYS fluent.

Approach

Normally we will use the dynamic mesh approach to derive this type of relations but due to over complexity and some lack of experience we choose a different approach by using moving reference frame "MRF" approach and some trials and errors to derive an approximate relation these trials and errors were done by fixing the mass flow rate and assuming the turbine RPM since we are working in a steady state we can tell that the total torque acting on the rotor is = ZERO

Steps

1. Generating geometry
2. Geometry preparation.
3. Mesh generation
4. Case setup
5. Solving

Computational Fluid Dynamics CFD

Create Geometry

First we need to create a simpler model for our device with less details to ease the upcoming processes

- 1) Make it a single body
- 2) Remove any unnecessary features
- 3) Make it all fillet and remove any sharp edges
- 4) Export the geometry as a “.step file”

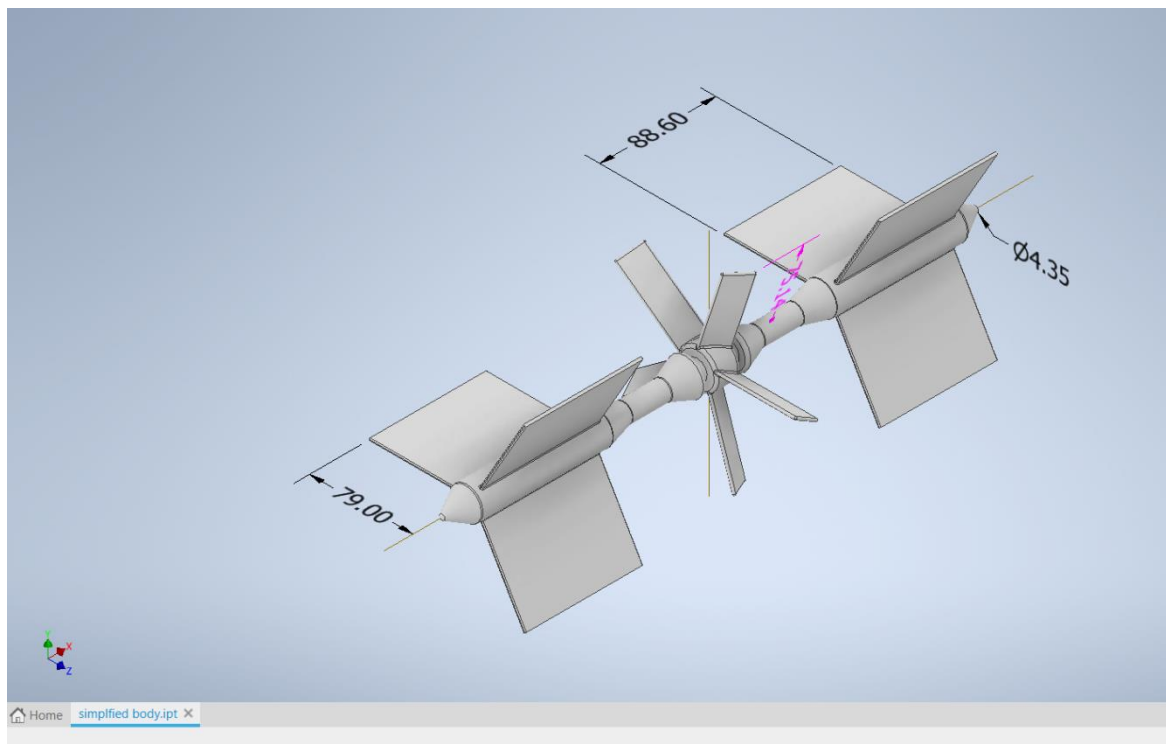


Figure 4 Geometry

Geometry preparation

We need to prepare the geometry for the meshing

So we insert our simplified geometry into ANSYS modeler “spaceclaim , design modeler”

Here we used space claim then we do some steps

- 1) Repair the geometry
- 2) Create the different fluid zones
- 3) Enclosure
- 4) MRF
- 5) Suppress the main body from physics and delete it
- 6) Start naming different faces and bodies for further steps

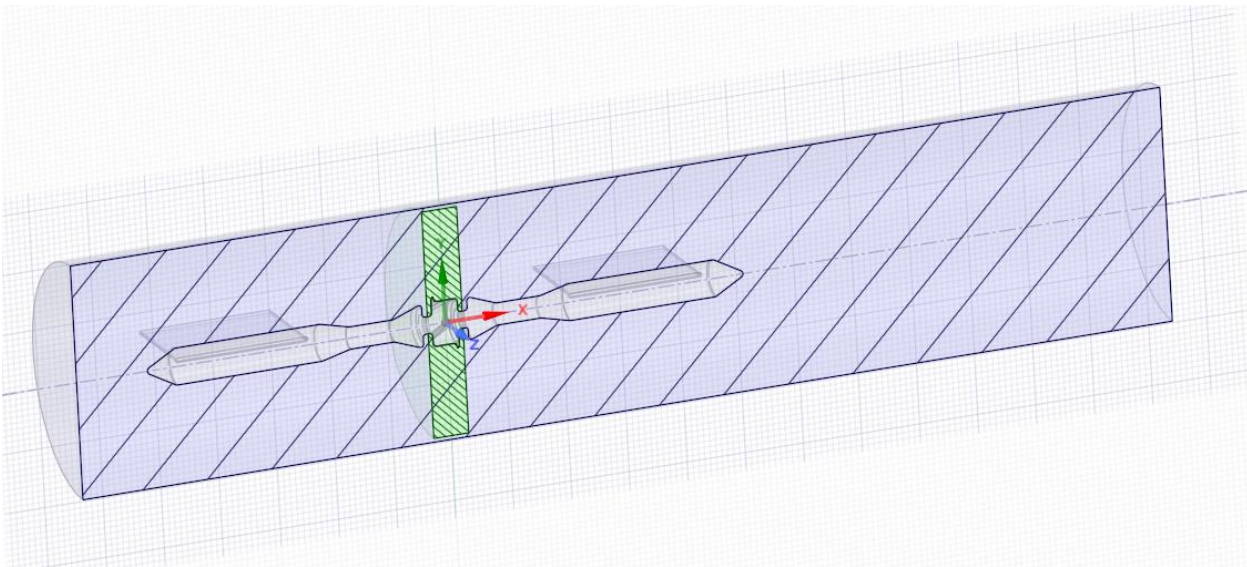


Figure 5 Fluid zones

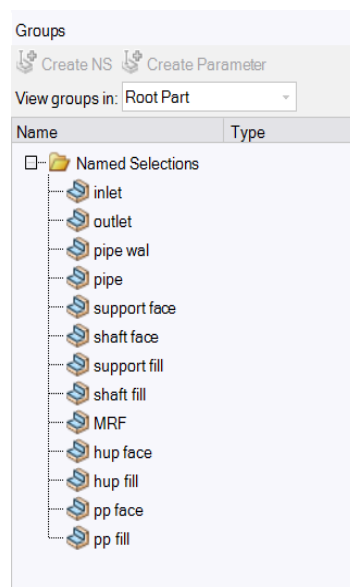


Figure 6 named selection

Meshing

The meshing is the most crucial part in any numerical analysis as it determine the accuracy of the results and whether it is solved or not so it must be done carefully and accurately

We tried meshing with 2 ways “design modeler, fluent mesher”

In the design modeler mesher there was a high skewness in the mesh and it quality was a pit poor although it had a large number of cells avg. 17mil cell

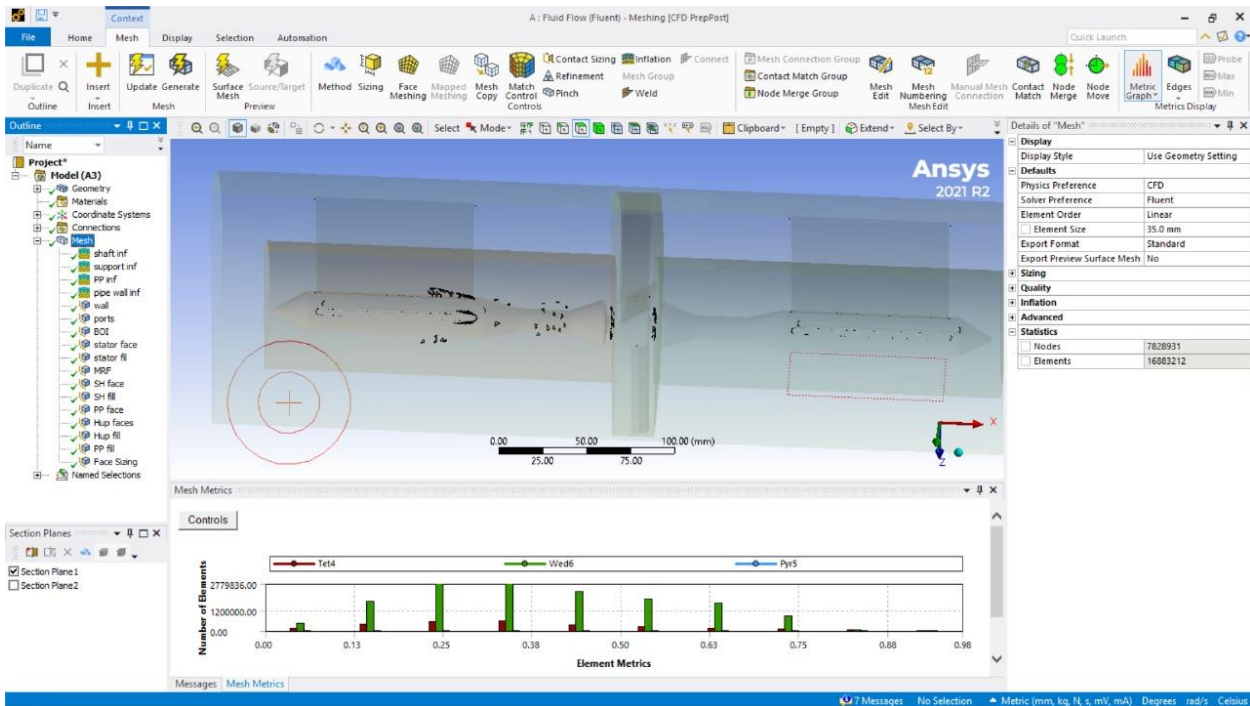


Figure 7 design modeler mesh

Due to this issues and warnings we turned to using fluent mesher which was simpler and more accurate

Meshing steps

- 1) Import geometry
- 2) Select local sizing to specify certain faces that I am more interested in studying or has a relatively small features to make its cell size small to improve the quality
- 3) Generate surface mesh and check its skewness
- 4) Specify the regions we have to only fluids
- 5) And inflation to the walls to make boundary layers

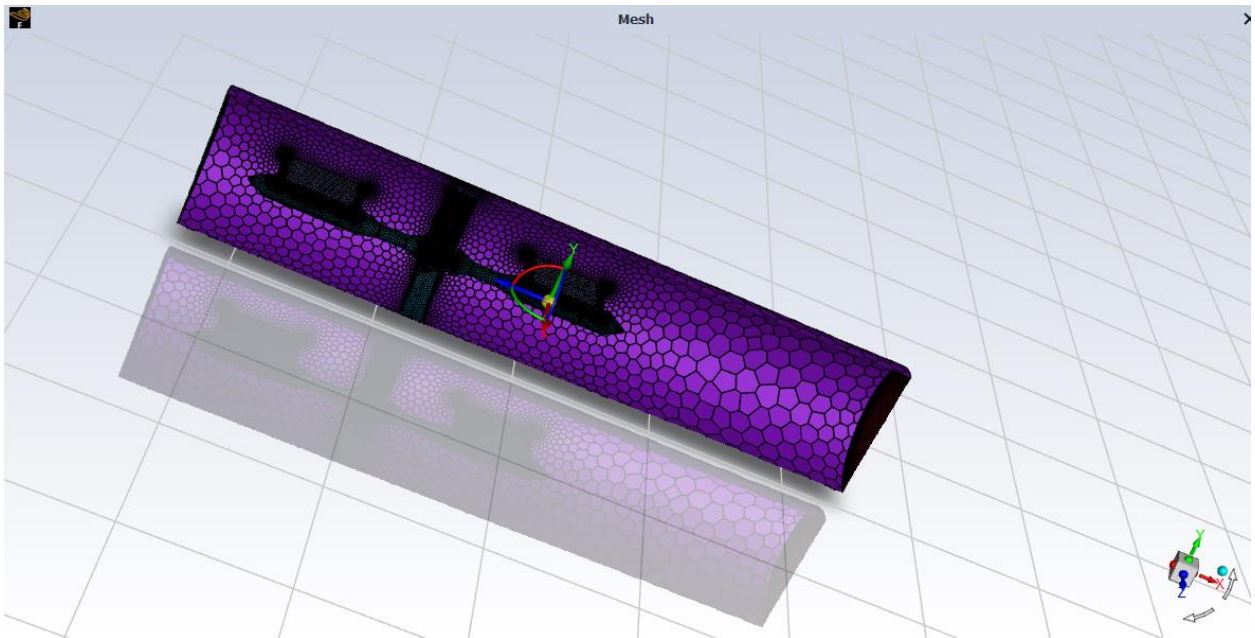


Figure 8 surface mesh

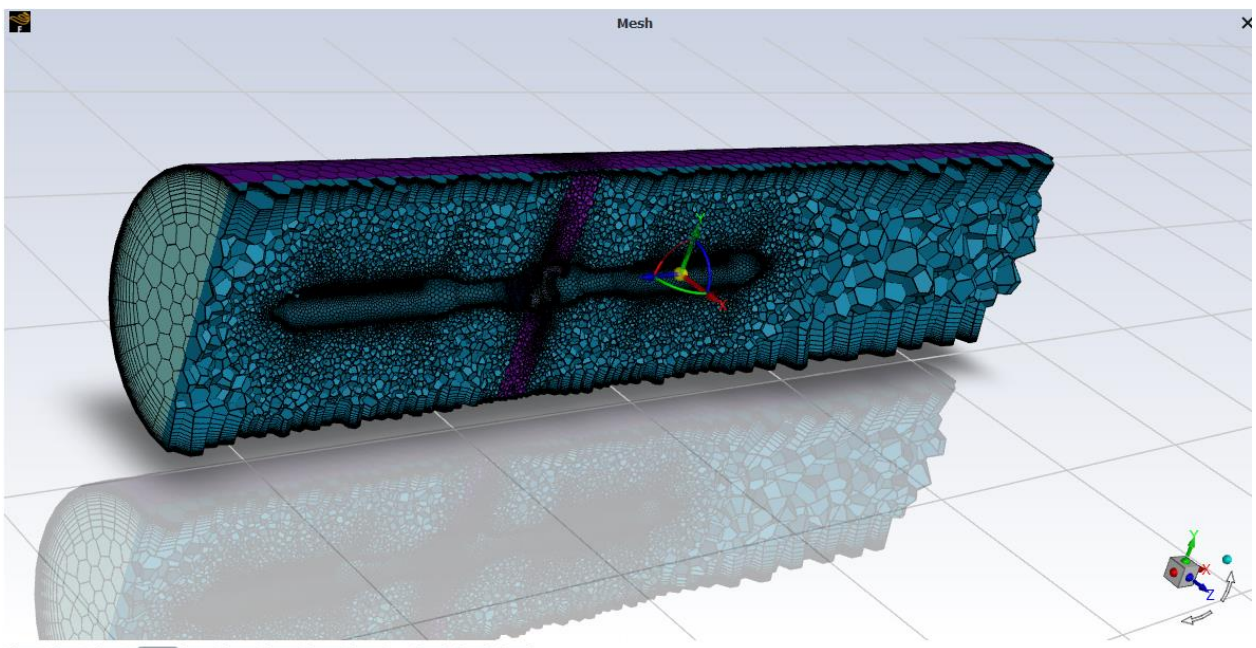


Figure 9 volume mesh

Casing

We the go to the solver and setup our case and add our boundary conditions :

1. Steady state pressure based solution
2. Using air as the fluid material
3. We have 2 fluid zones the pipe and “MRF”
4. We work on a flow rate inlet “which we will change in the trials”
5. Pressure outlet “ zero gauge pressure at outlet”
6. We set the MRF as a frame motion with rotational motion “changed in the trials ”

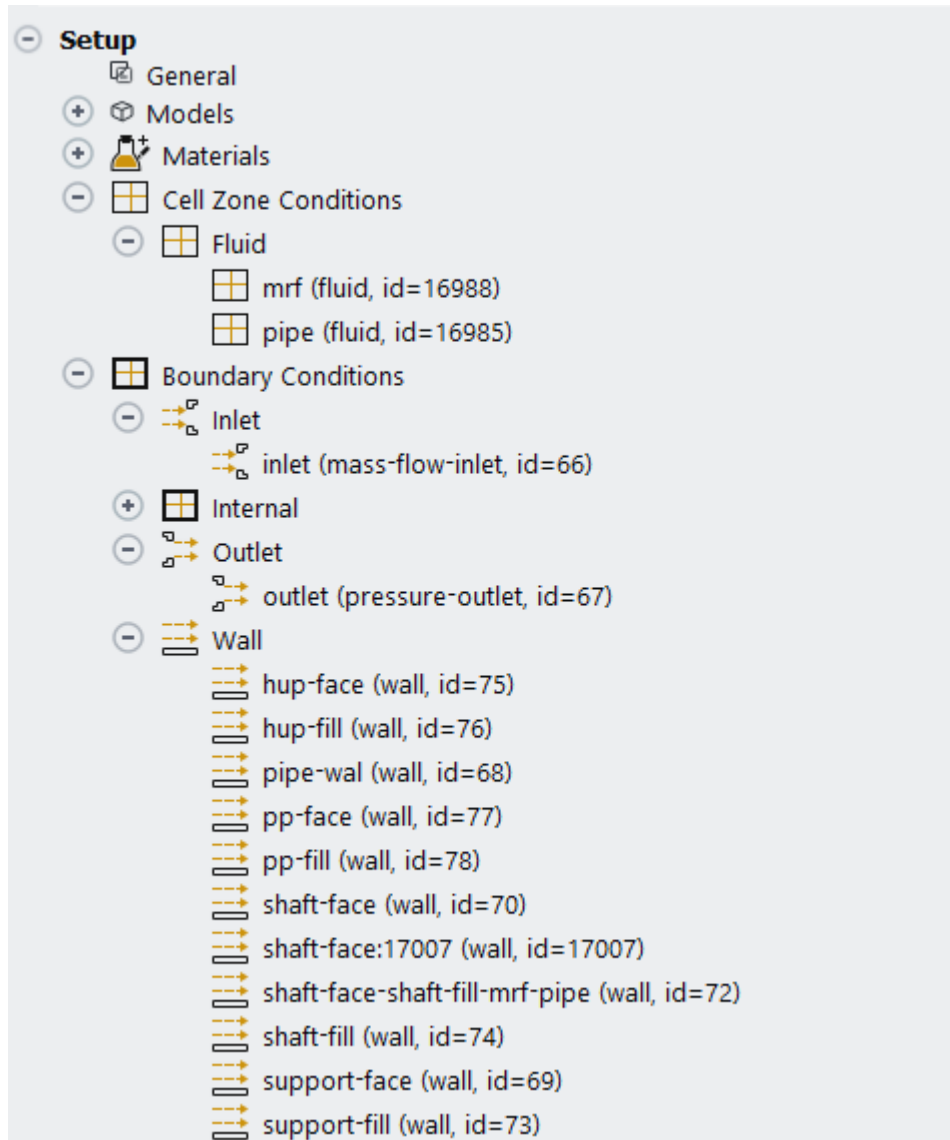


Figure 10 case setup

solution

we did some trials and errors for deferent mass flow rates and rpm and added a plot for the torque and observe it and try to equivalate to zero

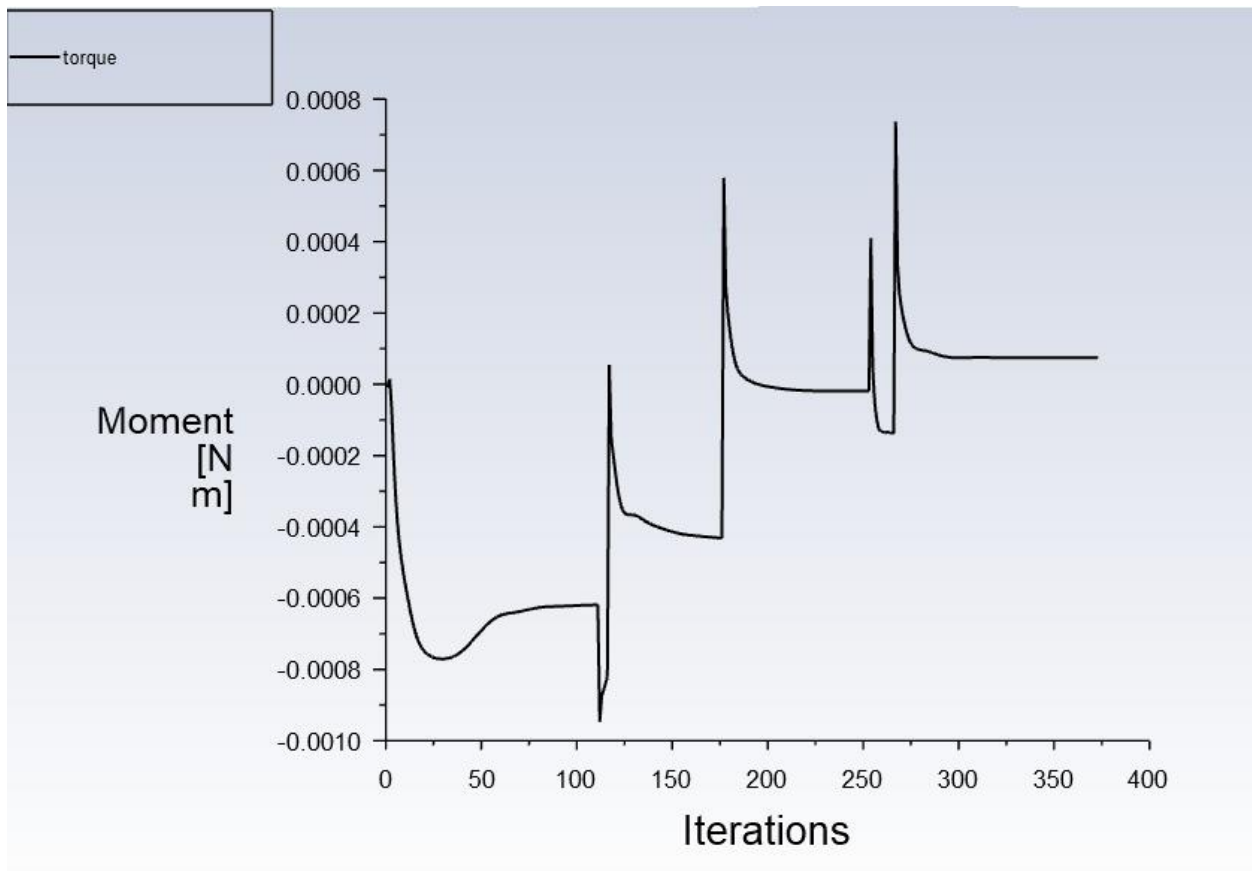
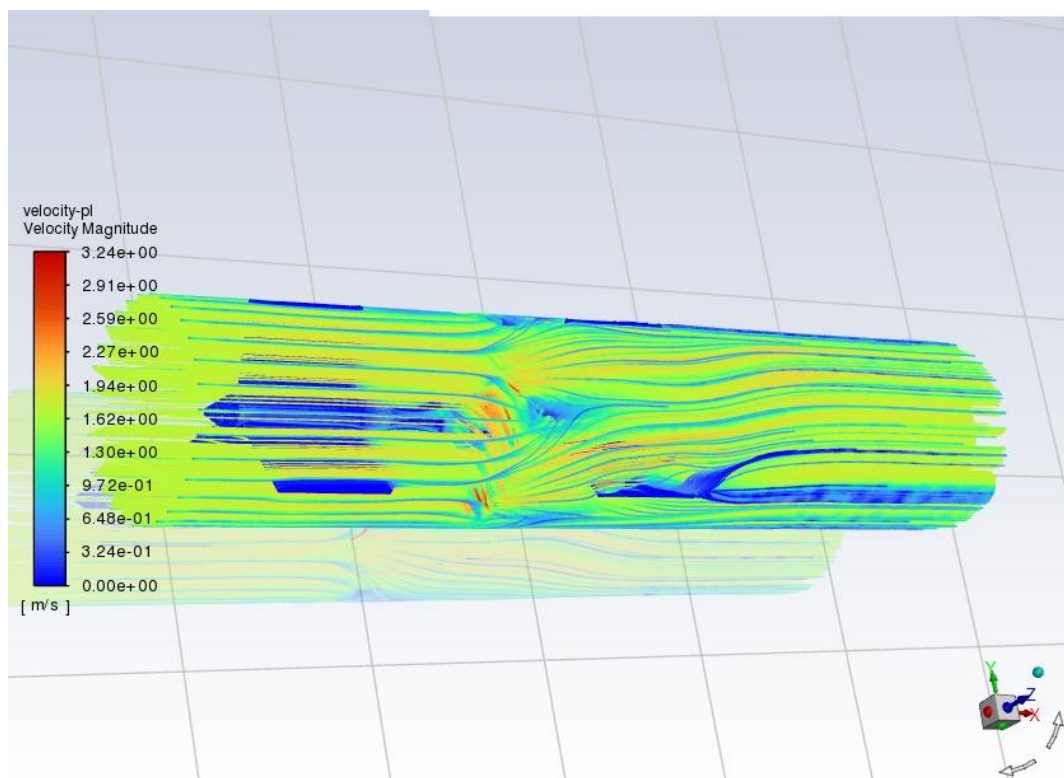


Figure 11 Torque iterations



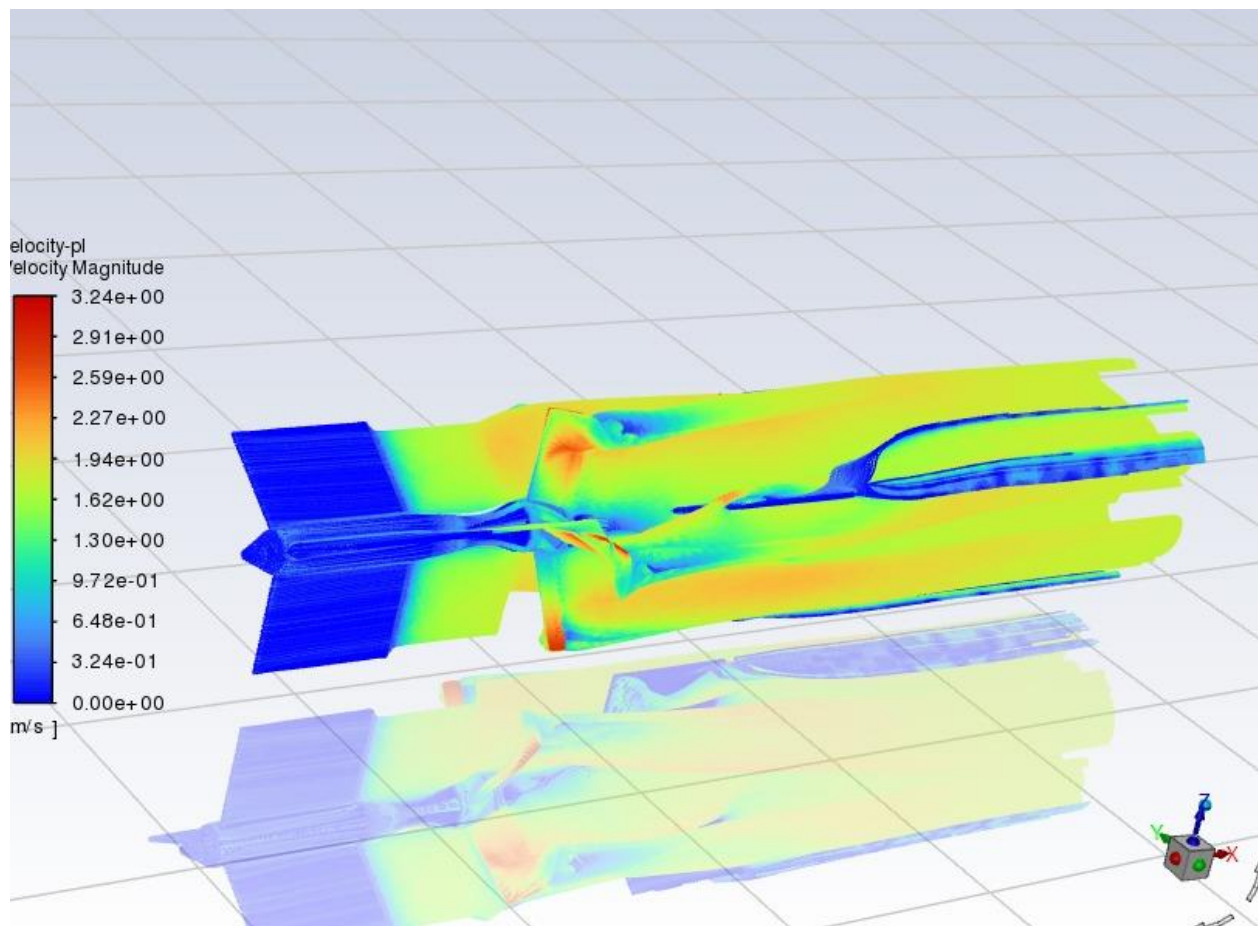


Figure 12 velocity path lines

Average of Facet Values Static Pressure		[Pa]
inlet		0.31010098
outlet		0
Net		0.15853217

Figure 13 pressure loss

Hanger Design

Upstream and Downstream Rotor Supports:

- ideally, these supports should have minimal impact on the fluid flow. Their design should aim to:
 - Minimize Form Drag: The supports should be streamlined to reduce pressure drag caused by flow separation and wake formation.
 - Avoid Flow Blockage: The supports shouldn't obstruct the flow path significantly, as this would reduce the mass flow rate through the turbine and decrease efficiency.
 - Prevent Upstream Disturbances: The upstream support should be designed to avoid creating turbulence or swirl in the flow *before* it reaches the rotor. This is crucial for optimal energy transfer.

Upstream and Downstream Cone:

these conical shapes likely serve a few important purposes related to fluid flow and structural integrity:

1. Gradual Flow Transition:

- Upstream Cone: The upstream cone acts as a nozzle, helping to gradually accelerate the incoming airflow towards the rotor. This smooth, conical transition minimizes flow separation and turbulence that would occur if the air jet suddenly encountered a flat surface or a sharp change in direction. A well-designed nozzle increases the kinetic energy of the air before it impacts the rotor, improving the turbine's efficiency.¹
- Downstream Cone: The downstream cone helps to gradually diffuse the airflow *after* it has interacted with the rotor. This smooth expansion reduces the kinetic energy of the exiting air, minimizing losses and potentially recovering some pressure. It also helps to guide the airflow away from the rotor supports and other components, reducing drag.

2. Improved Flow Alignment:

- Upstream Cone: The conical shape helps to align the incoming airflow with the rotor blades. This ensures that the air jet impacts the rotor at the optimal angle for maximum momentum transfer.
- Downstream Cone: The downstream cone helps to guide the exiting airflow smoothly away from the rotor, preventing it from interfering with the incoming air or creating back pressure.

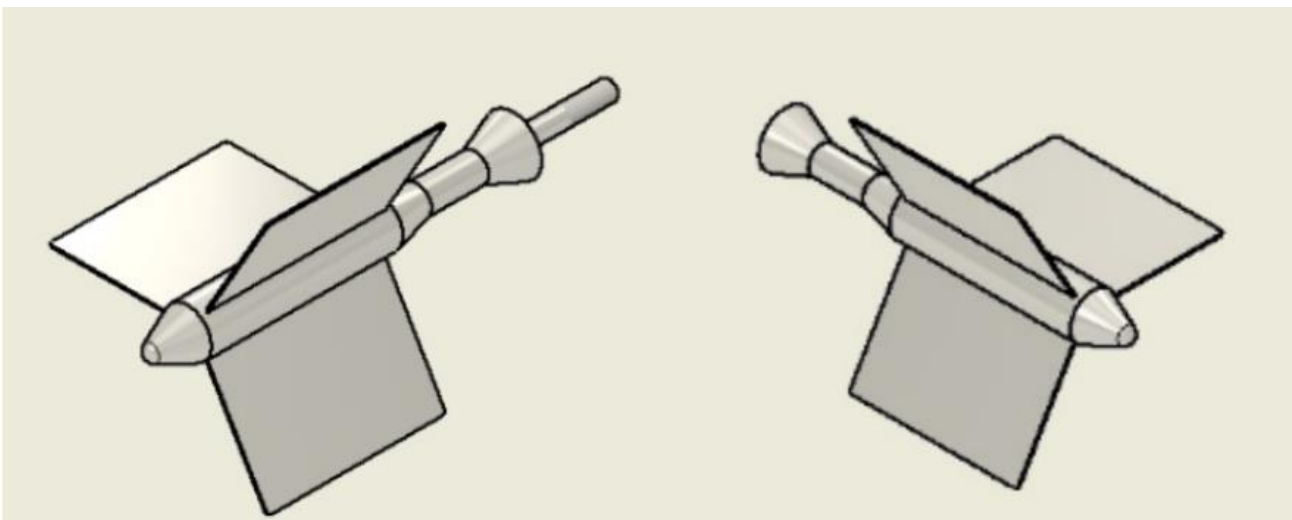


Figure 14 Upstream & Downstream Hungers

Rotor Design :

Research for

- Optimum blade length $\approx 0.7-0.9$ of Pipe Diameter
- Attack angle Range $\approx 15 - 30$ Degree
- Relation between pressure drop and both angle of attack and area of blade: that pressure drop increases when angle of attack or area of blade increases.
- Best blade shape for sensing (Straight Blade)
- Tangential Force generates RPM
- Torque = $F_t \times 0.45 \text{ Blade Length} + R_{hub}$
- $\mu = 0.2$
- $Re = \frac{\rho V D}{\mu}$
- $w = (5\% \rightarrow 10\%)$ of pipe Diameter , assumed to be 10%
- Lift Force “FL” $\propto$ Lift Coefficient “CL”
- Drag Force “FD” $\propto$ Drag Coefficient “CD”
- Cl & CD Depend on blade shape in range (CD= 0.8-1.3 , Cl=0.7-1.3)

Givens:-

- $P = 100.2 \text{ KPa}$
- $T = 298 \text{ K}$
- $R = 287 \text{ Pa} \cdot \text{m}^3/\text{Kg}$
- $\dot{M} = 0.04 \text{ Kg/sec}$
- $A = \frac{\pi}{4} D_{\text{pipe}}^2 = 0.00124 \text{ m}^2$
- $D_{\text{pipe}} = 6.25 \text{ inch} = 15.8 \text{ cm}$
- $R_{hub} = 1.5 \text{ cm}$ (for ball bearing 6800)
- From CFD $\rightarrow Cl = 1.1$, $Cd = 1.2$

Assumptions:-

- No. of blades (n) = 5
- Attack Angle (θ) = 30 Degree (Most Common)
- Frictionless motion (Friction Torque ≈ 0)
- Friction Torque $\propto \omega^2$

Rotor Calculations:-

Pre Calculations:-

- $D_{\text{rotor}} = 15.8 \times 0.9 = 14 \text{ cm}$
- $\rho = \frac{P}{RT} = 1.17 \text{ Kg/m}^3$
- $V = \frac{M^o}{\rho A} = 1.87 \text{ m/sec}$
- Surface Area = Blade Width(w) x Length(R) = 11 cm^2
- $t = 2 \text{ mm}$ Max(Assumption)
- $R = \frac{D_{\text{rotor}}}{2} - R_{\text{hub}} = 7 - 1.5 = 5.5 \text{ cm}$
- $w = 0.1 \times 15.8 = 1.58 \text{ cm} \approx 2 \text{ cm}$
- Projected Area of blade = $A \sin \theta = 11 \times \sin 30 = 5.5 \text{ cm}^2$

Forces on Blades

- $\sum F_x = \text{Tangential Force "Ft"}$
- $F_L \sin \theta + F_D \cos \theta = F_t$
- $\sum F_y = \text{Axial Force "Fa"}$
- $F_L \cos \theta - F_D \sin \theta = F_a$
- $F_D = C_d \frac{1}{2} \rho n A V^2 = 0.00586962 \text{ N}$
- $F_L = C_l \frac{1}{2} \rho n A V^2 = 0.00538 \text{ N}$
 - $\therefore F_a = 0.00777 \text{ N}$
 - $\therefore F_t = 0.00172 \text{ N}$

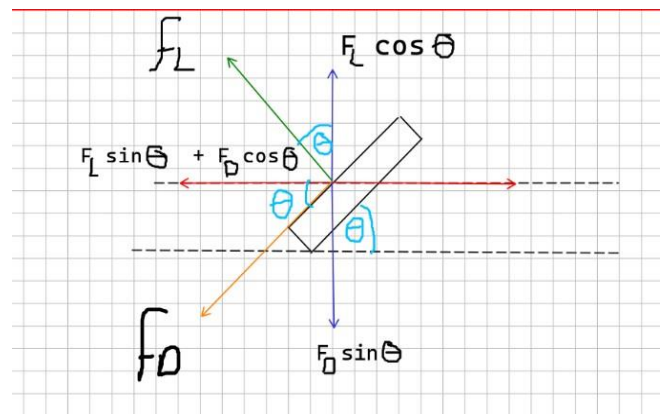


Figure 15 Forces on Blade

Torque Calculation:-

- $T = Ft(0.45 R + R_{hub}) = 6.85619 \times 10^{-5} \text{ N.m}$
- $T_{friction} = K \omega^2$
- Resultant Torque $= T - T_{friction} = 0$ after certain period

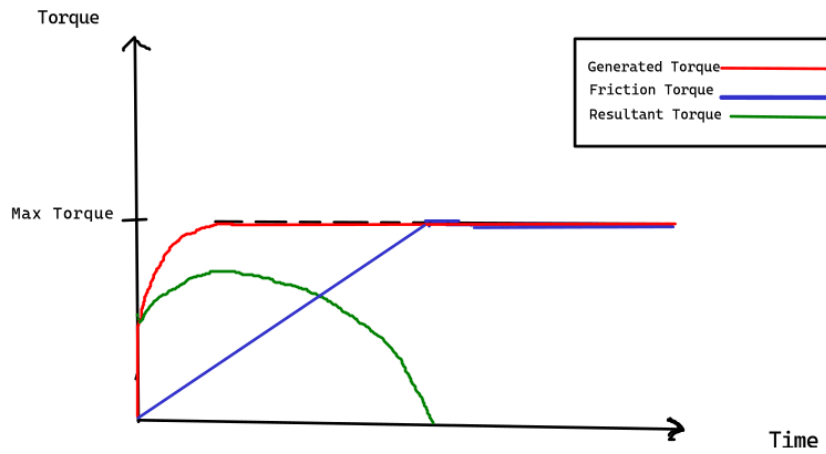


Figure 16 Torque with time

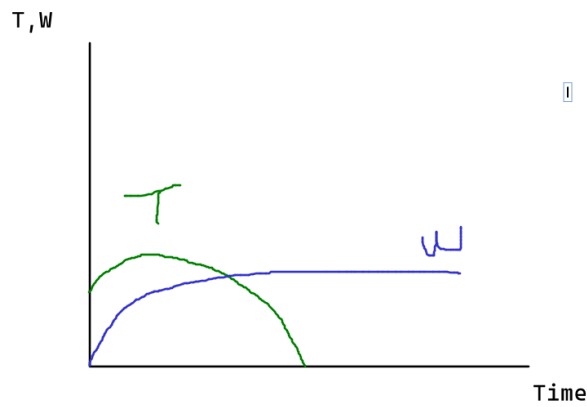


Figure 17 Torque with angular acceleration

Angular Acceleration (α):-

- $I = 0.00008 \text{ Kg.m}^2$
- $\alpha = \frac{T_{res}}{I} = 0.857023 \text{ rad/s}^2$

Angular velocity:-

- For Transient State $\rightarrow \omega = \alpha t$
- For Steady State $\rightarrow \omega = \text{constant}$

Mechanical Design:-

Flow Meter Assembly

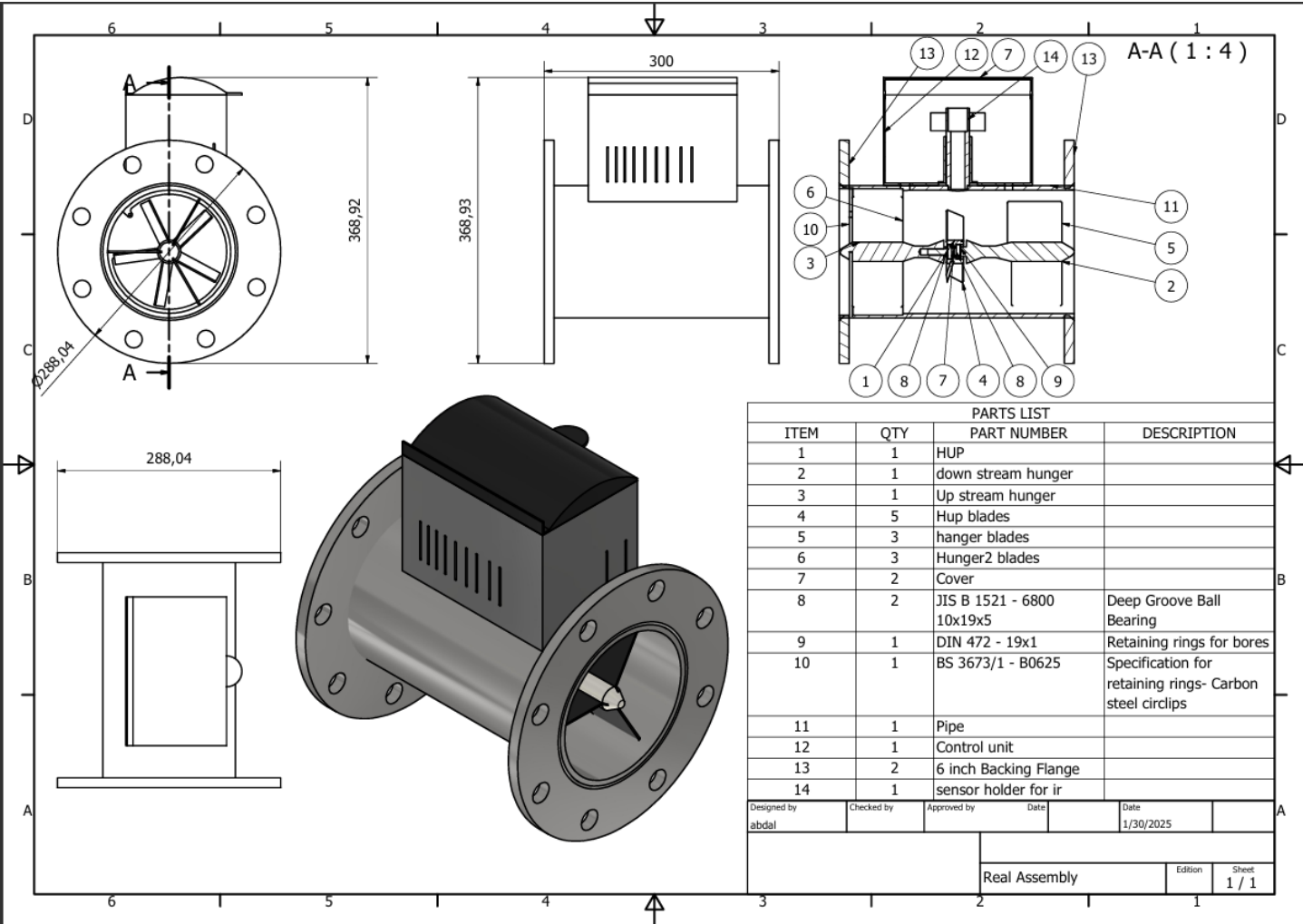


Figure 18 Working Drawing of Flowmeter Assembly

1) Housing

1-Pipe

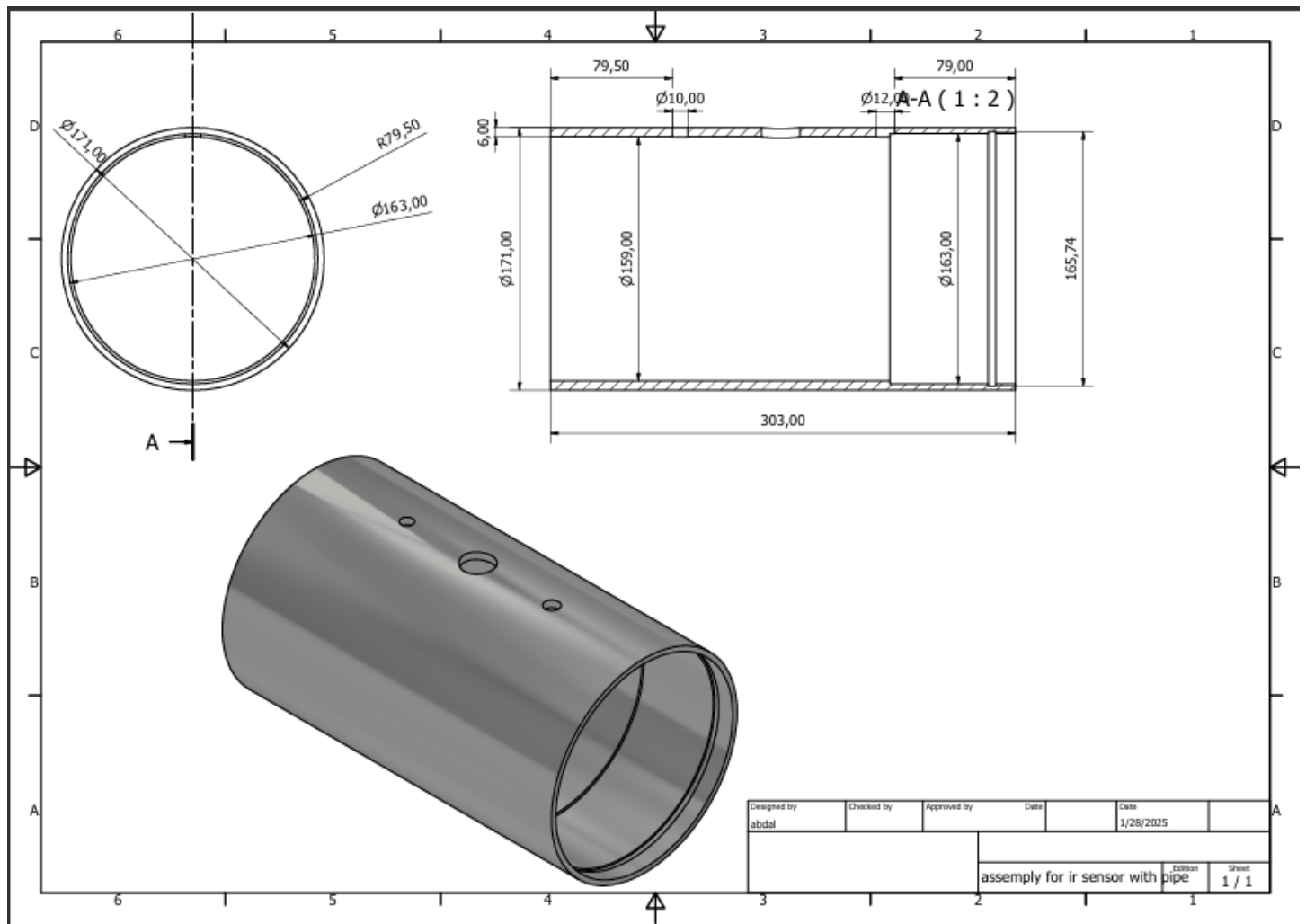


Figure 19 Working Drawing of Pipe

2-Flanges

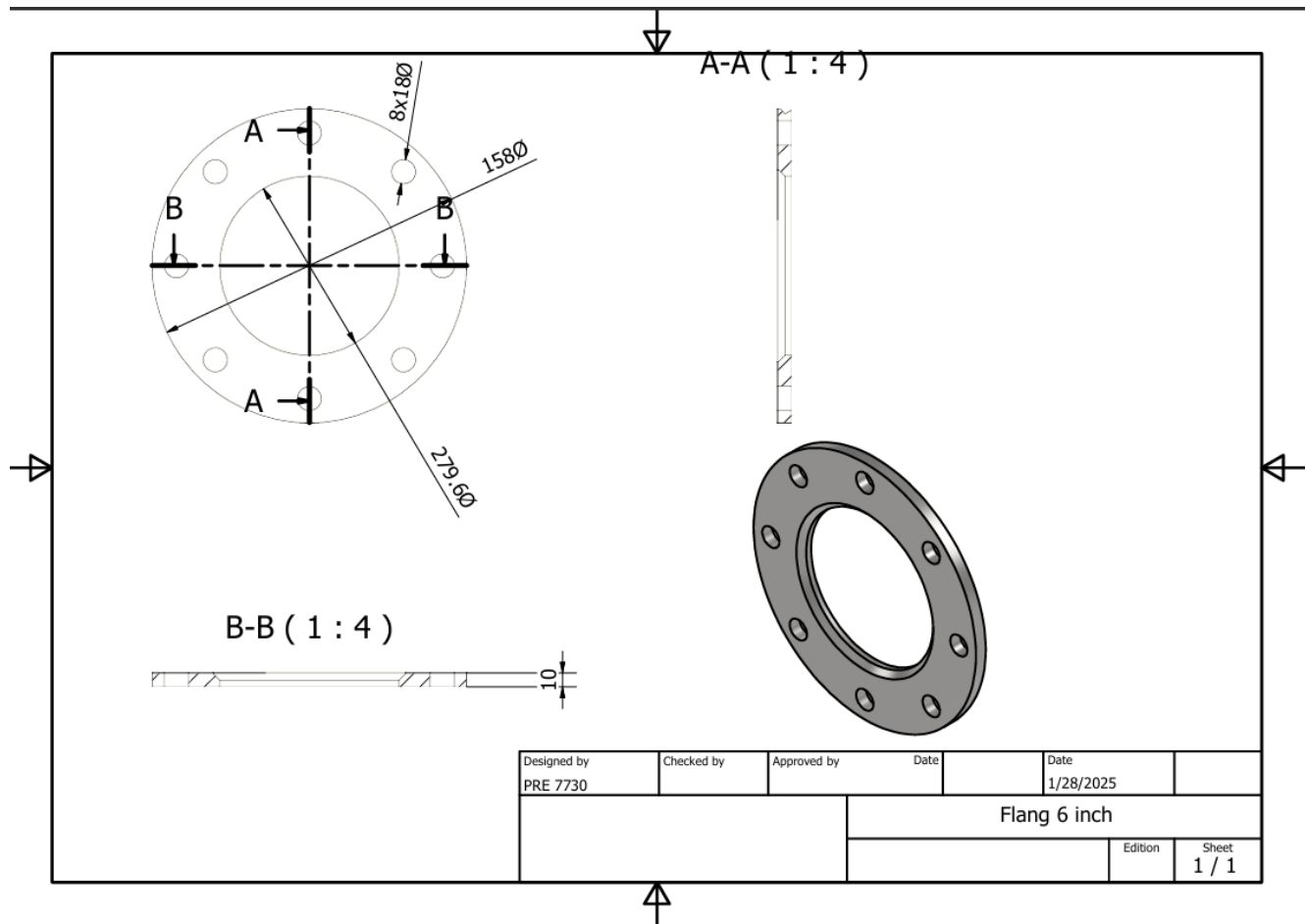


Figure 20 Working Drawing of Flange 6 inch ,8 holes

3- Electronics Holder

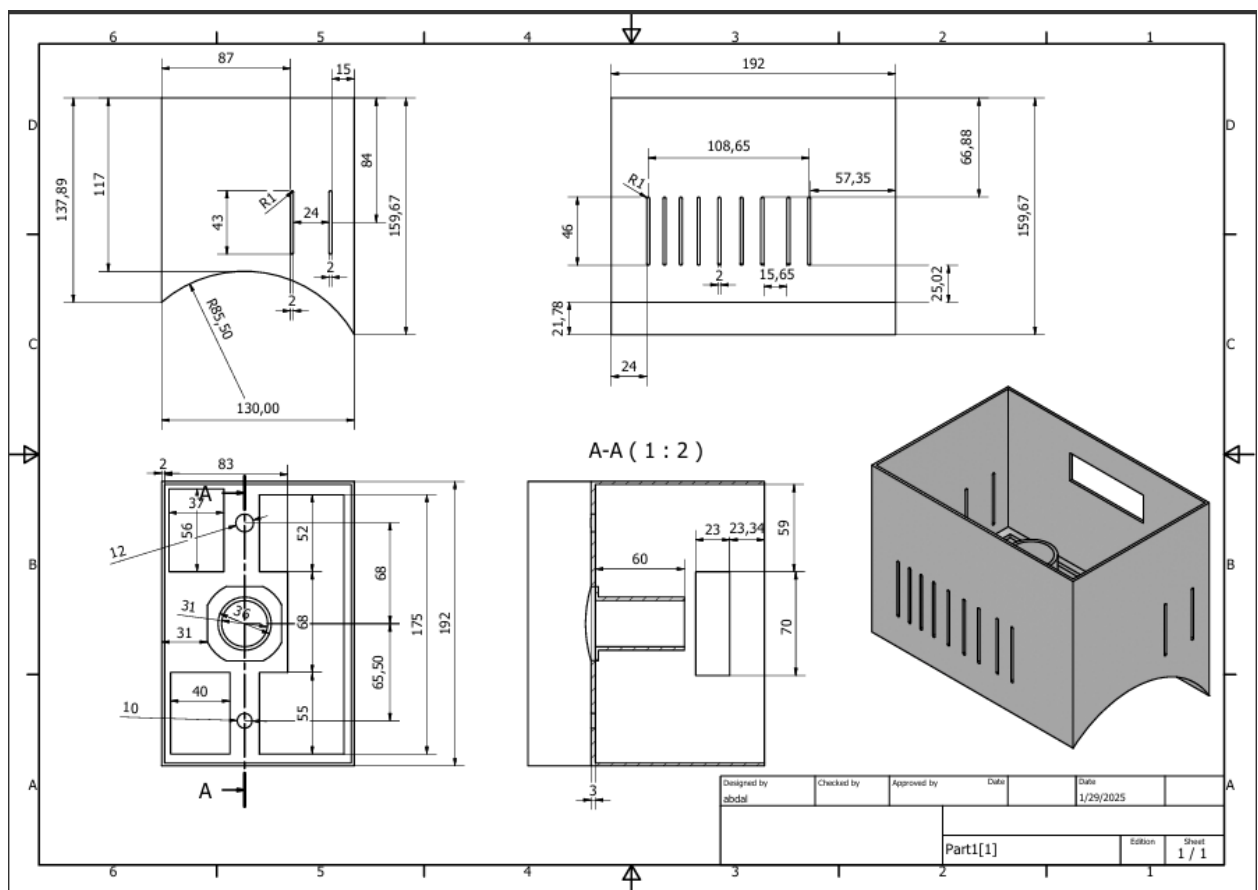


Figure 21 Working Drawing Of Electronic Holder

4-Sensor Holder

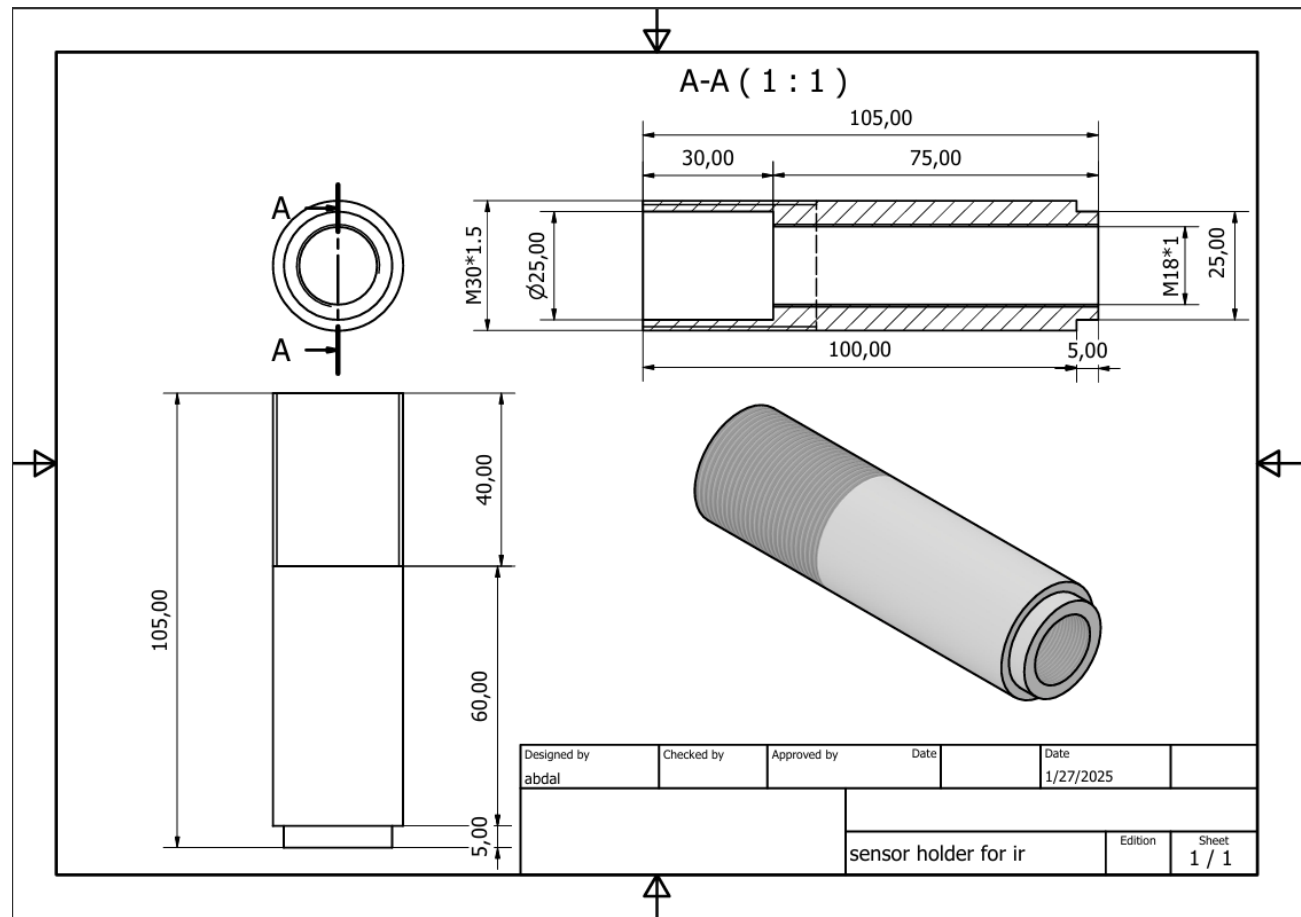


Figure 22 Working Drawing of Sensor Holder

4)Cover of Electronics Holder

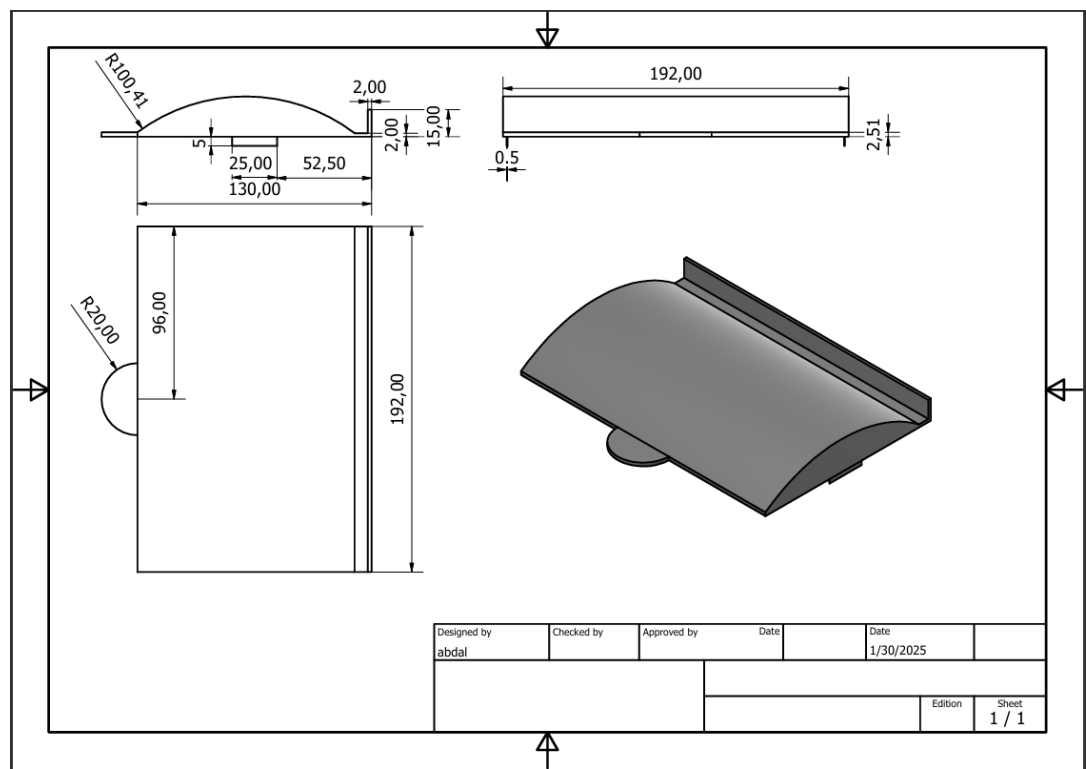


Figure 23 Working Drawing of Cover

pipe with holder Assembly:-

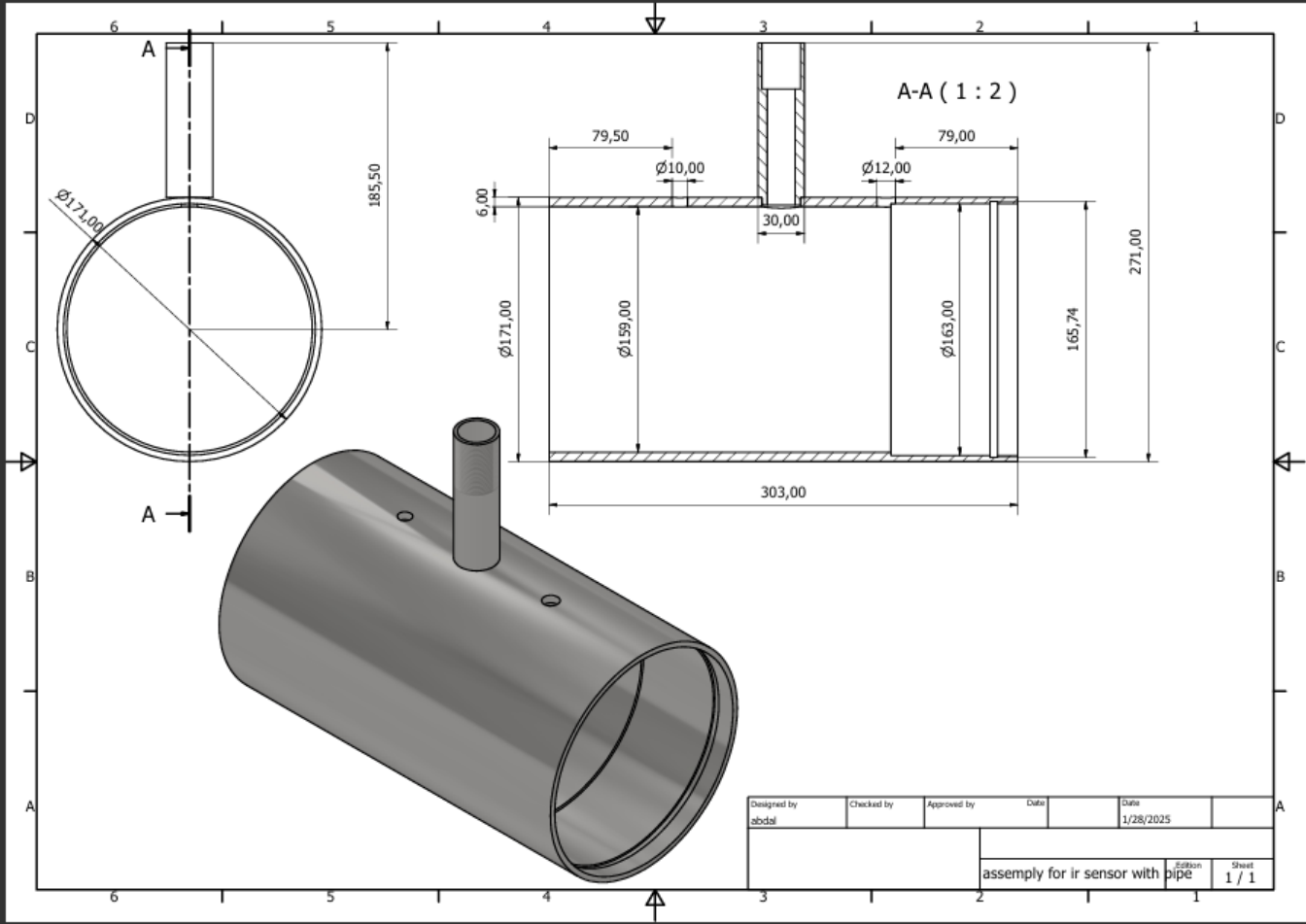


Figure 24 pipe with holder working Drawing

2) Hanger

1- Upstream Hanger

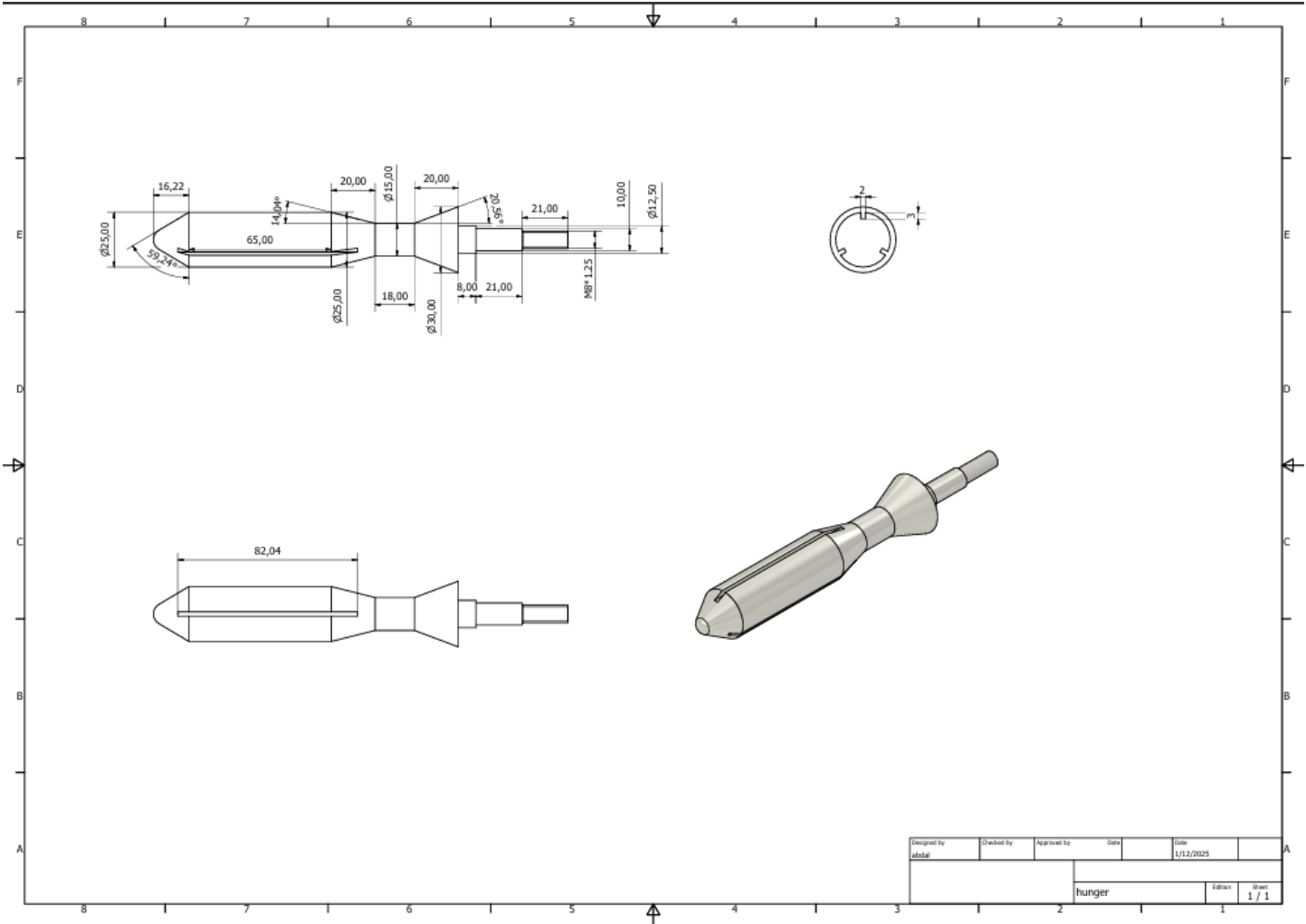


Figure 25 Working Drawing of Upstream Hanger

2- Upstream Blades

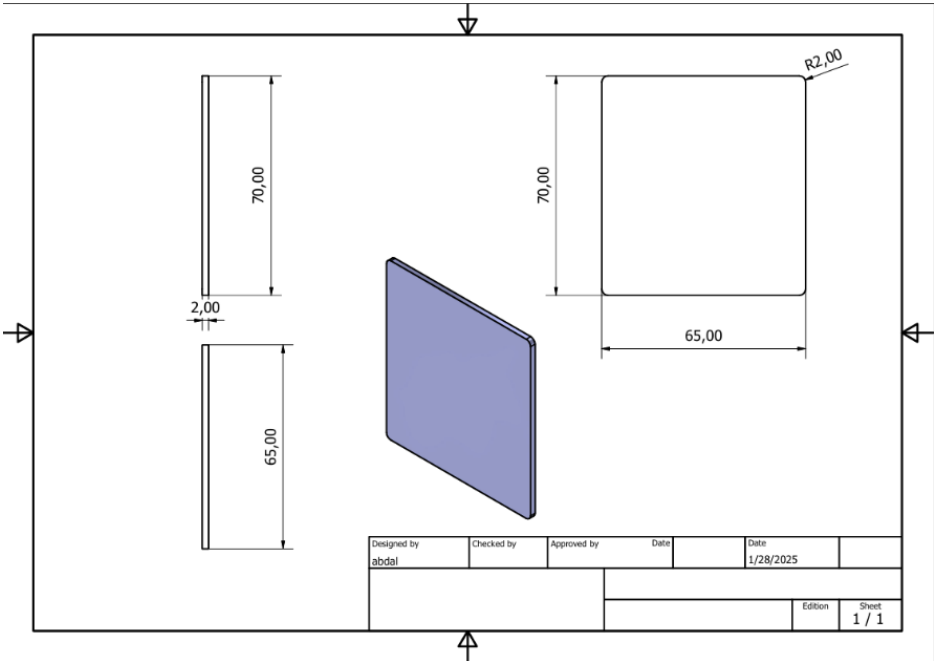


Figure 26 working drawing of Upstream Hanger

3- Downstream Hanger

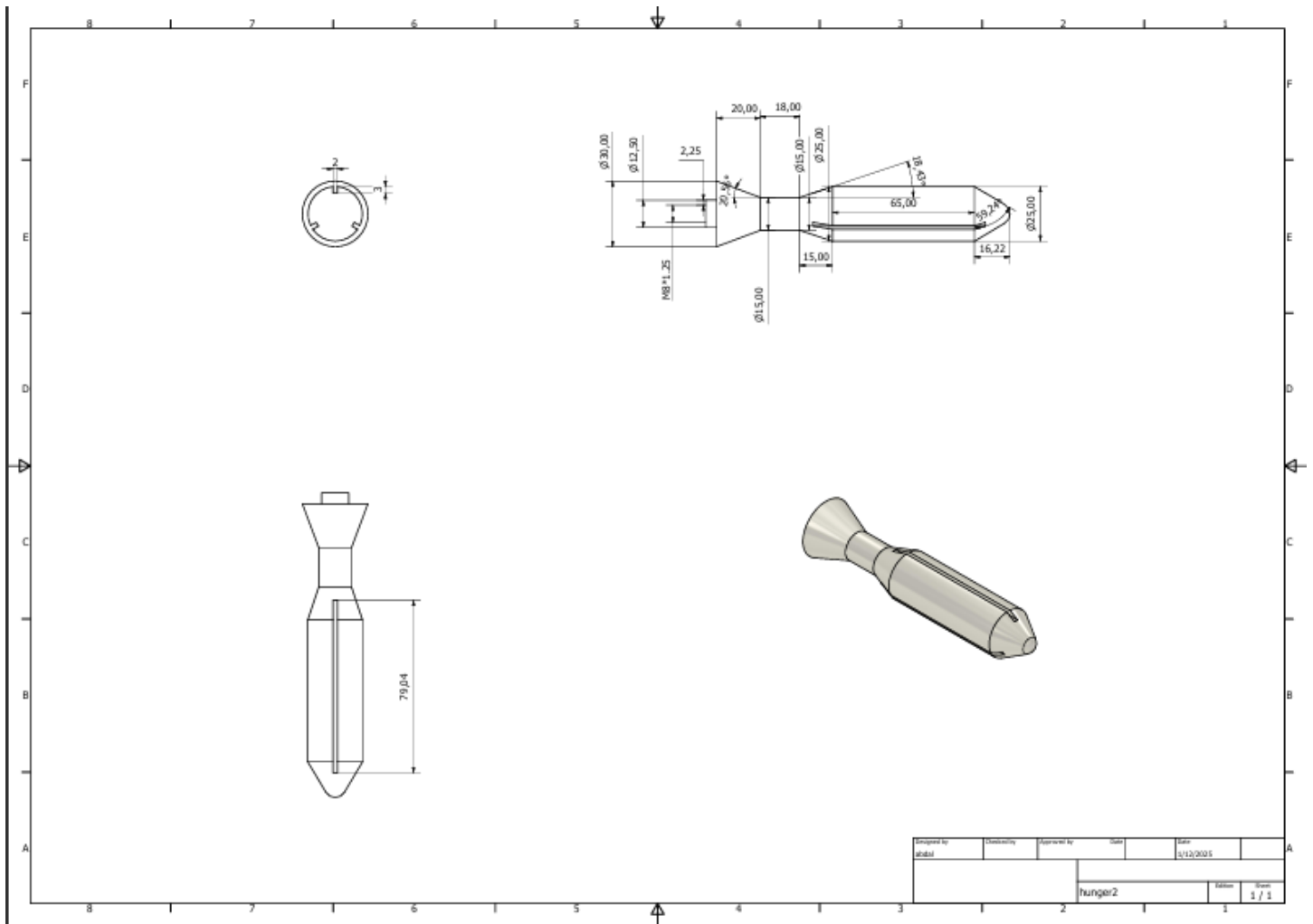


Figure 27 Figure 4 Working Drawing of Downstream Hanger

4- Downstream blades

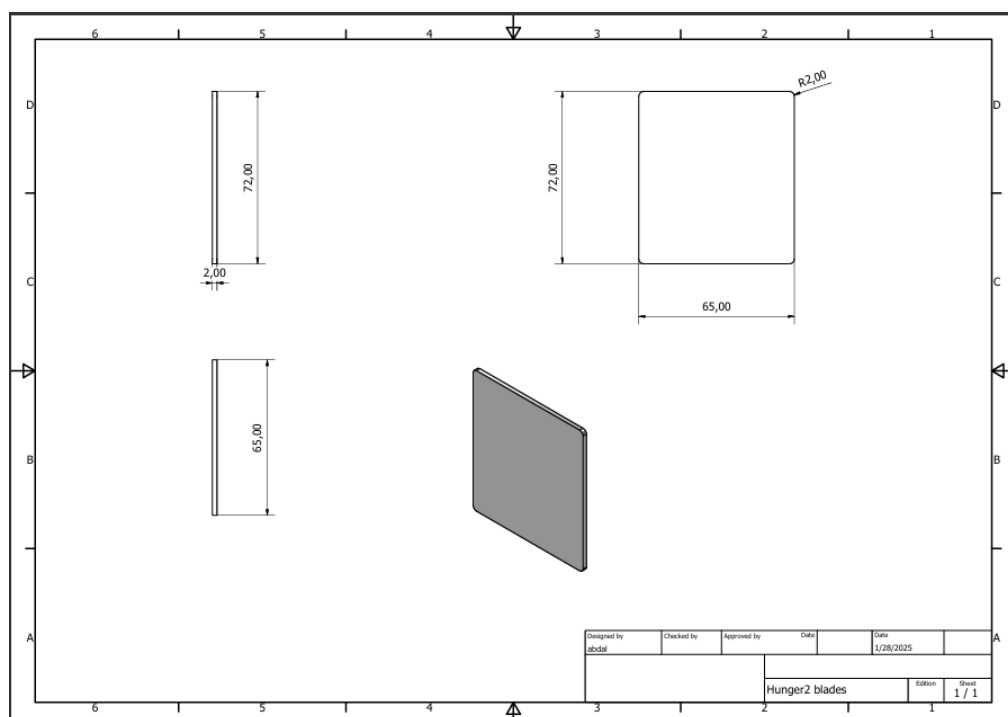


Figure 28 Working Drawing of Upstream Hanger

3) Rotor

1) Hub

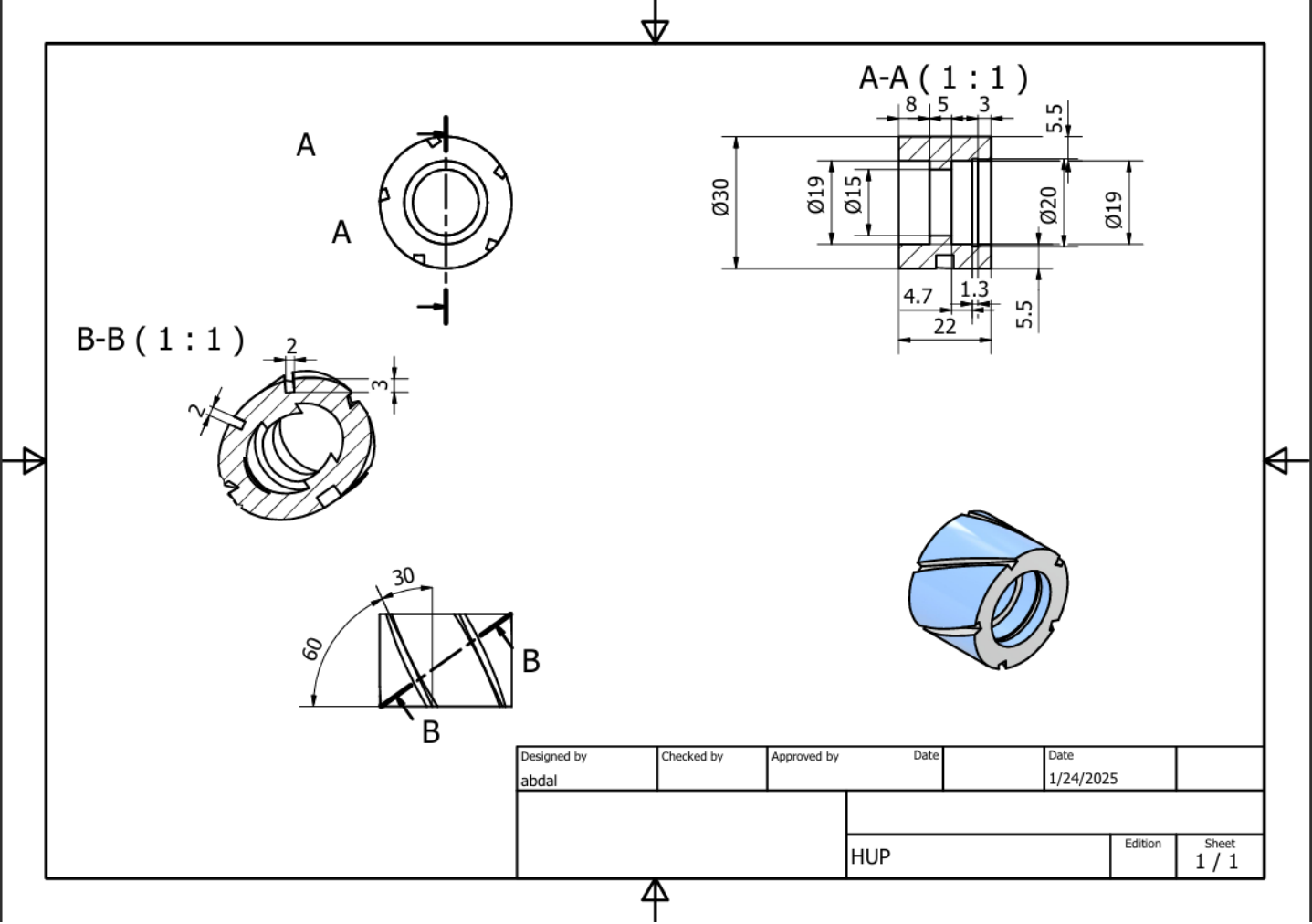


Figure 29 Working Drawing of Hub

2) Rotor Blades

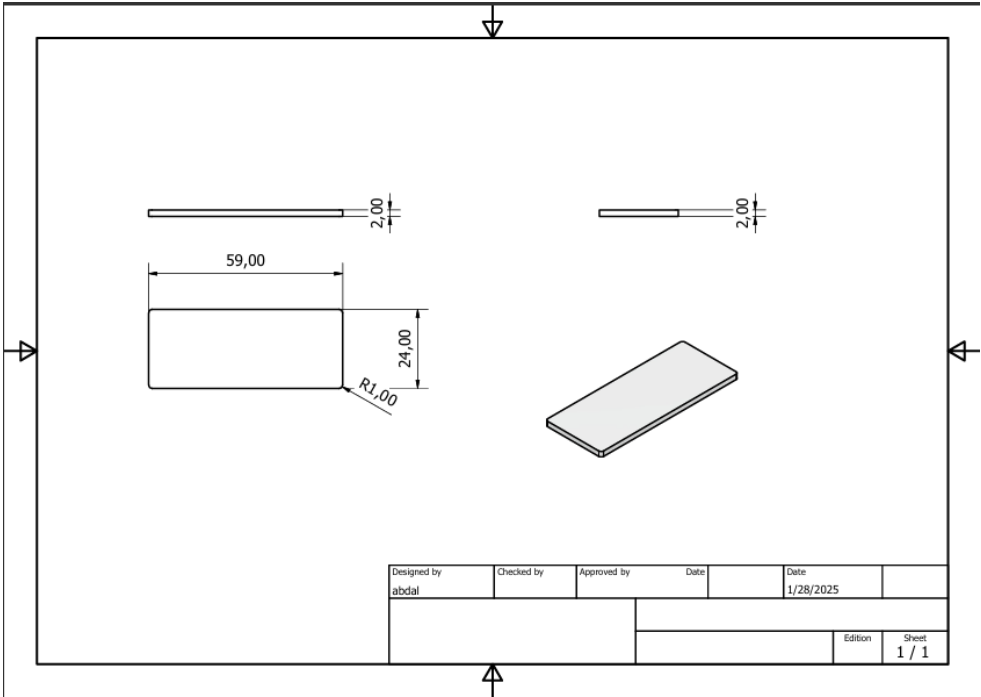


Figure 30 Rotor Blades

3) Spacer

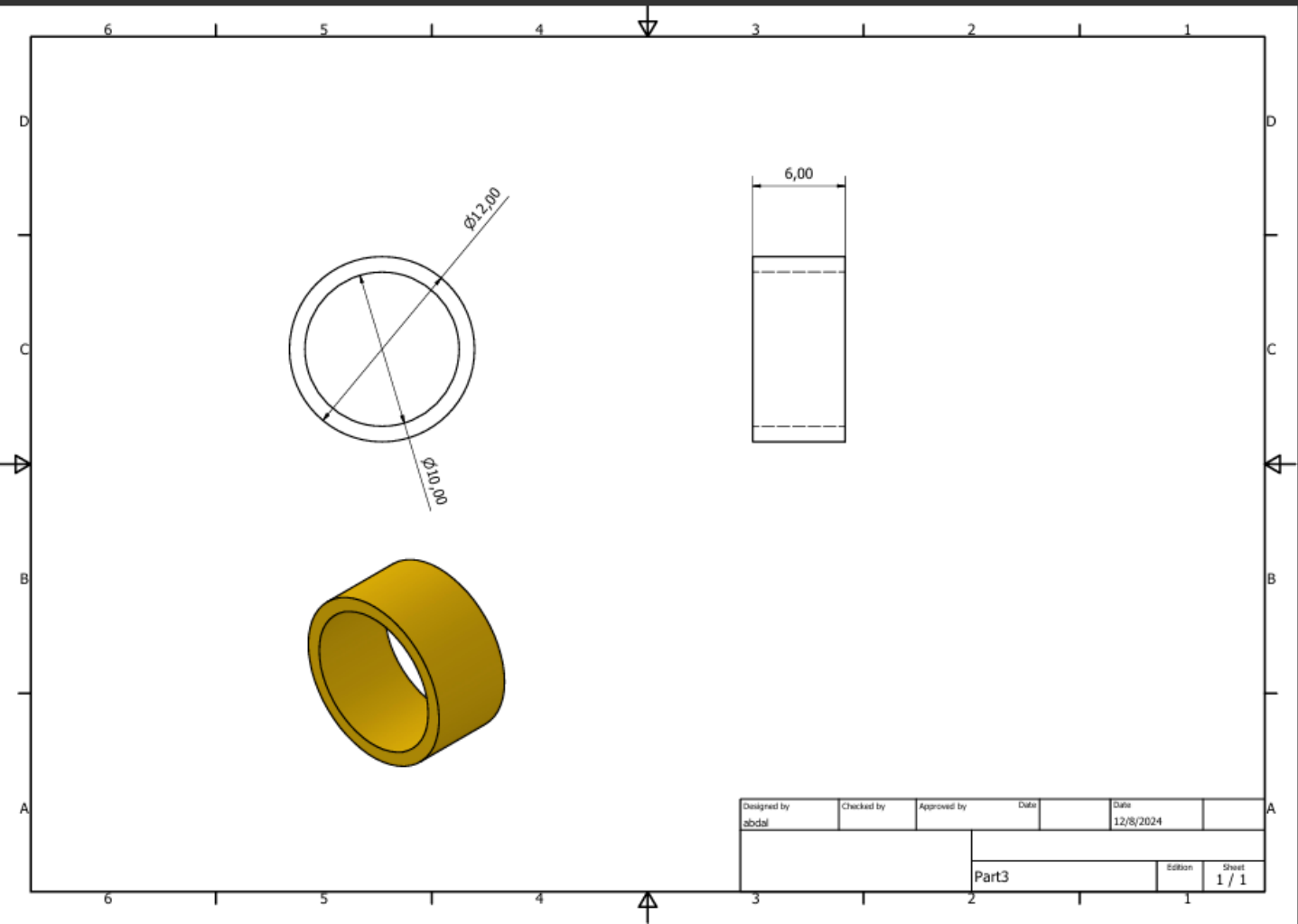


Figure 31 Working Drawing of Spacer

Hanger with Rotor Assembly

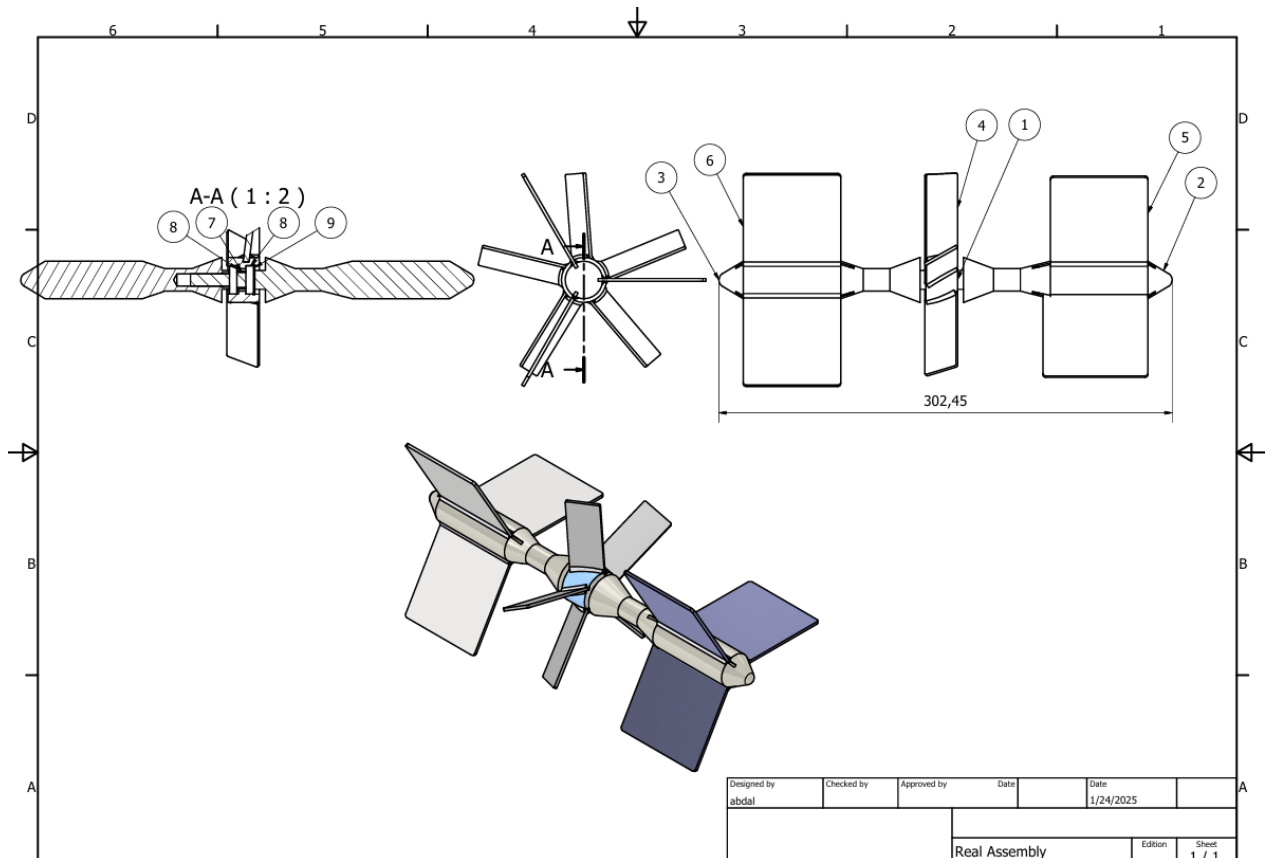


Figure 32 Hanger assembly Working Drawing

4) Standard Parts

- 1- Snap Ring for fixation of bearing 19(Eternal)
- 2- Snap Ring for fixation of Hanger 160(Eternal)
- 3- 2 deep groove ball Bearing 6800
- 4- Joint
- 5- Bolts M16
- 6- Nuts M16
- 7- Washer
- 8- Gasket 6 INCH,8 holes

How we Manufactured the Flowmeter:-

Manufacturing of st37 Rod

Description: it is a long rod we separated it to 3 parts one for sensor holder with thread that holds the IR sensor above pipe perpendicular to the rotating rotor and two identical parts for stator shaft .

- **Material:** **Steel 37**
- **Raw material dimension:** **700 × Ø32**

Machine used:

- 1- Turning Machine

No. of operations: **13**

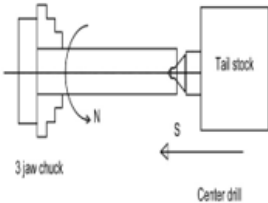
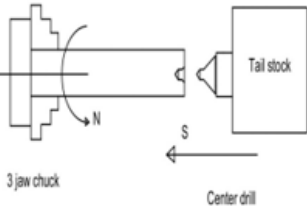
Fixation tools:

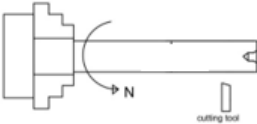
- 1-Three jaw chuck
- 2-live center

Cutting tools:

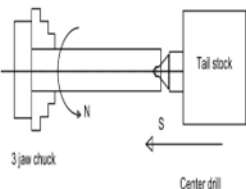
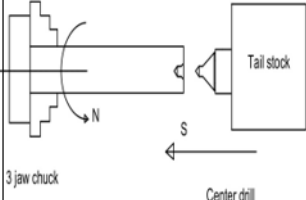
- 1-single cutting tool (turning)
- 2-Parting off tool
- 3-Internal thread tool

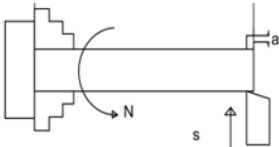
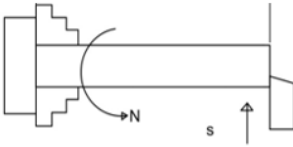
❖ **Total machining time is 30.981 min.**

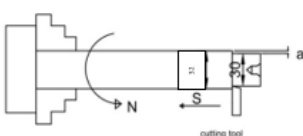
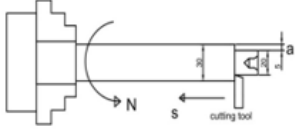
<u>Operation</u>	<u>Description</u>		<u>Cutting condition</u>
1-centering	Before 	After 	Feed = 0.1 mm/rev -width of cut = 2.5mm -Run in = 2mm - length=5mm - N = 200 m/min. - no of strokes = 1 $T=2+5/200*0.1=0.14 \text{ min}$
	Remarks - fixation : center drill is fixed in tail stock Workpiece is fixed by 3 jaw chuck - width of cut = 5mm (the length of tool tip) - objective : the center drill opens a small hole for the dead center in order to fix the W.P as it is a little bit long W.P and to continue all the following turning operations without bending concern.		

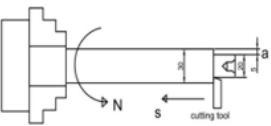
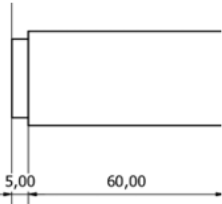
Operation	Description		Cutting condition
2- parting off	Before 	After 	Feed = 0.1 mm/Rev -width of cut =6mm -Length=17 mm -Run in = 2mm -Run out = 2mm - N = 350 rpm - no of strokes = 3 - $T=3*(17+2+2)/350*0.1=1.625 \text{ min}$
	Remarks - fixation : 3-jaw chuck - width of cut : 6 mm (the parting off cutter width) - objective : cutting off the whole part to separate the W.P to three parts with the wanted dimensions.		

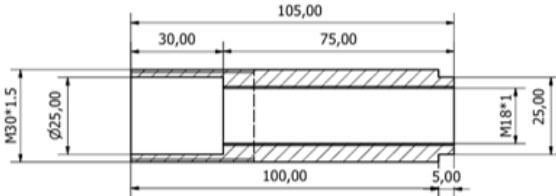
For Sensor Holder:

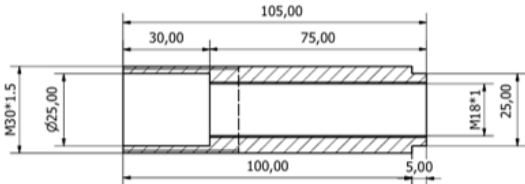
Operation	Description		Cutting condition
3-centering	Before 	After 	Feed = 0.1 mm/rev -width of cut = 2.5mm -Run in = 2mm - length=5mm - N = 200 m/min. - no of strokes = 1 $T = 2 + 5/200 * 0.1 = 0.14 \text{ min}$
	Remarks - fixation : center drill is fixed in tail stock Workpiece is fixed by 3 jaw chuck - width of cut = 5mm (the length of tool tip) - objective : the center drill opens a small hole for the dead center in order to fix the W.P as it is a little bit long W.P and to continue all the following turning operations without bending concern.		

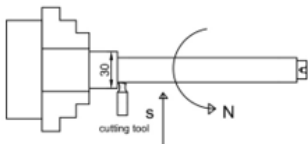
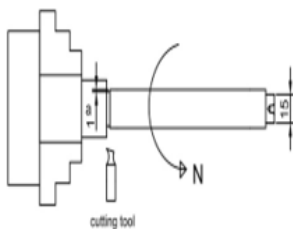
Operation	Description		Cutting condition
4-Facing	Before  cutting tool	After 	Feed = 0.16 mm/Rev -depth of cut = 2mm -Run in = 2mm length = 17mm - N = 500 - no of strokes = 1 -T = 17.5 + 2 + 2/500 * 0.16 = 0.3 min
	Remarks *Fixation : 3-jaw chuck *Depth of cut : 2mm *Objective: flattening the surface and assure part dimension. commonly is done as the first machining operation.		

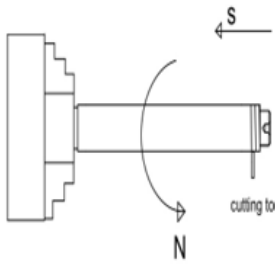
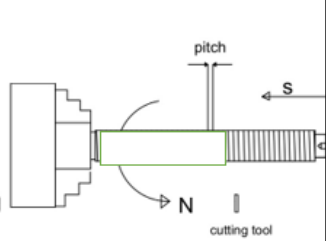
<u>Operation</u>	<u>Description</u>		<u>Cutting condition</u>
5- longitudinal turning (Ø32 to Ø30)	Before 	After 	Feed = 0.2 mm/Rev -depth of cut = 1 mm -length = 105 mm -Run in = 2mm - N = 400 ,500 rpm - no of strokes = $\frac{ma}{a} = 2$ $T = (105 + \frac{1}{400} \times 0.3) = 1.48 \text{ min}$
	Remarks -fixation: 3 jaw chuck and live center -Machining allowance = $d_i - d_f / 2 = 32 - 30 / 2 = 1 \text{ mm}$ *depth of cut per first stroke = 2 mm /stroke *and second stroke= 0.5mm/stroke a=1 i=2		

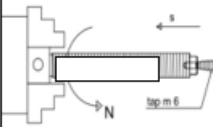
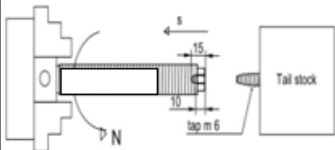
Operation	Description		Cutting condition
6- longitudinal turning (Ø30 to Ø25)	Before 	After 	Feed = 0.3 ,0.2 mm/Rev -depth of cut = 2,2,1 mm -length = 5 mm -Run in = 2mm - N = 400 rpm - no of strokes = $\frac{ma}{a} = 3$
	Remarks -fixation: 3 jaw chuck and live centers -machining allowance = 5mm *depth of cut per first stroke = 2 mm /stroke *And second stroke = 1mm/stroke - a=5 - i=2 - Objective: achieving the surface of Ø 25 which will be threaded later		$T = (5 + \frac{2}{400} \times 0.3) + (5 + \frac{2}{400} \times 0.3) + (5 + \frac{1}{400} \times 0.3) = 1.4 \text{ min}$

Operation	Description		Cutting condition
7- Drilling Ø25	Before	After	Feed = 0.1 mm/rev. - Length = 30mm. -Run in = 2mm. - N =500 rpm. - No. of strokes = 1. -T/stroke = (12.5+2)/500* 0.1 = 0.34 min
	Remarks Fixation: 3-jaw chuck Depth of cut: 12.5mm. (width of cutter) Objectives: To open a hole for entering the sensor to be fixed on the M18 thread. 		

Operation	Description		Cutting condition
8- Drilling Ø18	Before	After	Feed = 0.1 mm/rev. - Length = 65mm. -Run in = 2mm. - N =500 rpm. - No. of strokes = 1. -T/stroke = $(9+2)*65/500*$ 0.1 = 0.76 min
	Remarks Fixation: 3-jaw chuck Depth of cut: 9mm. (width of cutter) Objectives: To open a hole for entering the sensor to be fixed on the M18 thread. 		

Operation	Description		Cutting condition
9- Under cutting On $\varnothing 30$	Before 	After 	- Feed = 0.1mm/rev. - Depth of cut = 1mm - Run in = 2mm. - Run out = 2mm. - N = 500 rpm. - No. of strokes = 1 - T = $(1+2+2)/500 \times 0.1$ = = 0.1 min
	Remarks - fixation: 3 jaw chuck and live center - ma = $30-28/2 = 1\text{mm}$ - Depth of cut = 1 mm - objective :we make this process on the end of threaded portion of screw to provide clearance for the cutting tool during the threading.		

Operation	Description		Cutting condition
10- External threading on $\varnothing 20$	Before 	After 	- Feed = 0.2mm/rev. - Depth of cut = 1mm - Length = 150mm - Run in = 2mm. - Run out = 2mm. - N = 300 rpm. - No. of strokes = 1. - T = $150+2+2/300*0.2 = 2.5 \text{ min}$
	Remarks Fixation:- 3-jaw chuck and live center Ma = $20-18/2 = 1 \text{ mm}$ Objectives: transfer the rotational motion to longitudinal motion to hold parts • The external thread can be either made manually by using a die , or automatically by fixing the tool on tool post and automatically powering our feed by lead screw. • -Feed is equal to the pitch of the thread = 0.2mm/rev MA = $(20-16)/2 = 2\text{mm}$.		

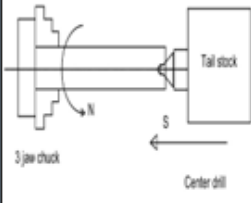
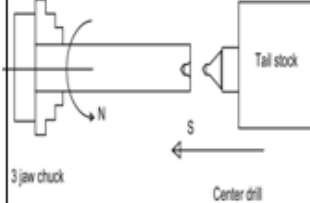
<u>Operation</u>	<u>Description</u>		<u>Cutting condition</u>
11- Internal thread Ø 17 to M18	Before 	After 	Feed = 0.1 mm/Rev -length = 10mm -Run in = 2mm - N = 300 - no of strokes = 1 T = (18+2/300*0.1)= 0.4 min
	Remarks - fixation : 3-jaw chuck and fixing the tool-on-tool post. - depth of cut : 0.5 mm - Note : automatically powering our feed by lead screw. **Feed is equal to the pitch of the thread=0.1mm*. - objectives : preparing for the M18 that will be fixed here.		

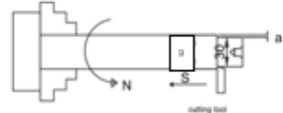
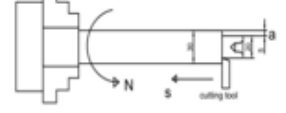
Final Product:-

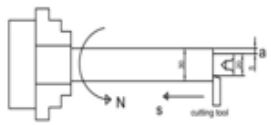
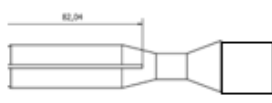
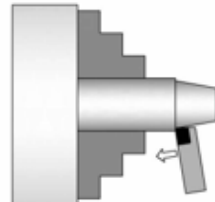


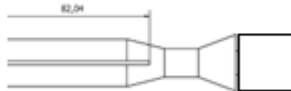
Figure 33 Sensor Holder

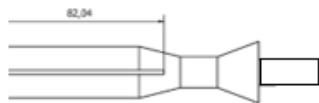
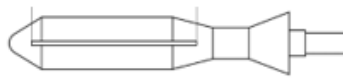
For Upstream and Downstream Hanger:*

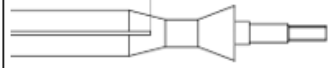
Operation	Description		Cutting condition
12-centering	Before 	After 	Feed = 0.1 mm/rev -width of cut = 2.5mm -Run in = 2mm - length=5mm - N = 200 m/min. - no of strokes = 1
	Remarks -fixation: center drill is fixed in tail stock Workpiece is fixed by 3 jaw chuck -width of cut = 5mm (the length of tool tip) - objective : the center drill opens a small hole for the dead center in order to fix the W.P as it is a little bit long W.P and to continue all the following turning operations without bending concern.		$T=2+5/200*0.1=0.14 \text{ min}$

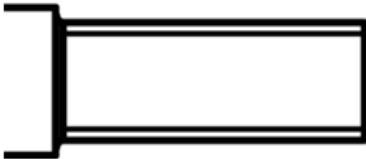
Operation	Description		Cutting condition
13- longitudinal turning (ø32 to ø25)	Before 	After 	Feed = 0.2 mm/Rev -depth of cut = 1 mm -length = 145mm -Run in = 2mm - N = 400 ,500 rpm - no of strokes = $m/a = 2$ $T = (145 + 1/400 * 0.3) = 1.64 \text{ min}$
	Remarks -fixation: 3 jaw chuck and live center -Machining allowance = $d_i - d_f / 2 = 32 - 25 / 2 = 3.5 \text{ mm}$ *depth of cut per first stroke = 2 mm /stroke *and second stroke= 0.5mm/stroke a=1 i=2		

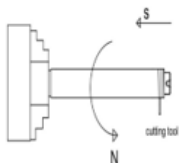
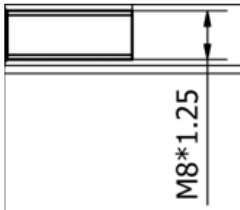
Operation	Description		Cutting condition
14- Tapered turning	Before 	After 	Feed = 0.2 mm/Rev -depth of cut = 1 mm -length = 145mm -Run in = 2mm - N = 400 ,500 rpm - no of strokes = $\frac{ma}{a} = 2$ $T = \frac{(145 + 1/400 * 0.3)}{.3} = 3.04\text{min}$
	Remarks -fixation: 3 jaw chuck and live center *depth of cut per first stroke = 2 mm /stroke *and second stroke= 0.5mm/stroke $a=1$ $i=2$ 		

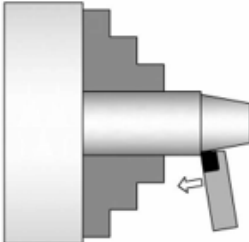
<u>Operation</u>	<u>Description</u>		<u>Cutting condition</u>
15- longitudi nal turning ($\varnothing 25$ to $\varnothing 12.5$)	Before 	After 	Feed = 0.2 mm/Rev -depth of cut = 1 mm -length = 50mm -Run in = 2mm - N = 400 ,500 rpm - no of strokes = ma/a = 4 T = 4*(8+1/400*0. 3)= 1.43min
	Remarks -fixation: 3 jaw chuck and live center -Machining allowance = $d_i-d_f/2= 25-12.5/2= 6.25$ mm		

<u>Operation</u>	<u>Description</u>		<u>Cutting condition</u>
16- longitudi nal turning (Ø12.5 to Ø10)	Before 	After 	Feed = 0.2 mm/Rev -depth of cut = 1 mm -length = 21mm -Run in = 2mm - N = 400 ,500 rpm - no of strokes = ma/a = 4 T = 1*(21+1/400* 0.3)= 0.32min
	Remarks -fixation: 3 jaw chuck and live center -Machining allowance =$d_i-d_f/2= 12.5-10/2= 1.25$ mm		

<u>Operation</u>	<u>Description</u>		<u>Cutting condition</u>
17- longitudi nal turning (Ø10 to Ø8)	Before 	After 	Feed = 0.2 mm/Rev -depth of cut = 1 mm -length = 21mm -Run in = 2mm - N = 400 ,500 rpm - no of strokes = ma/a = 4 T = 1*(21+1/400* 0.3)= 0.32min
	Remarks -fixation: 3 jaw chuck and live center -Machining allowance = $d_i-d_f/2= 10-8/2= 1$ mm		

Operation	Description		Cutting condition
18- Under cutting On $\varnothing 8$	Before	After	- Feed = 0.1mm/rev. - Depth of cut = 1mm - Run in = 2mm. - Run out = 2mm. - N = 500 rpm. - No. of strokes = 1 - T = $(1+2+2)/500 \times 0.1 = 0.1 \text{ min}$
	 Remarks -fixation: 3 jaw chuck and live center -ma = $30-28/2 = 1\text{mm}$ -Depth of cut = 1 mm - objective :we make this process on the end of threaded portion of screw to provide clearance for the cutting tool during the threading.		

Operation	Description		Cutting condition
19- External threading on $\varnothing 20$	Before 	After 	- Feed = 0.2mm/rev. - Depth of cut = 1mm - Length = 21mm - Run in = 2mm. - Run out = 2mm. - N = 300 rpm. - No. of strokes = 1. - T = $21+2+2/300 \times 0.2 = 0.416 \text{ min}$
	Remarks Fixation:- 3-jaw chuck and live center Objectives: -Feed is equal to the pitch of the thread = 0.2mm/rev		

<u>Operation</u>	<u>Description</u>		<u>Cutting condition</u>
20- Tapered turning	Before 	After 	Feed = 0.2 mm/Rev -depth of cut =1 mm -length =145mm -Run in = 2mm - N = 400 ,500 rpm - no of strokes = $ma/a = 2$ T = (145+1/400*0 .3) = 3.04min
	Remarks -fixation: 3 jaw chuck *depth of cut per first stroke =2 mm /stroke *and second stroke= 0.5mm/stroke $a=1$ $i=2$ 		

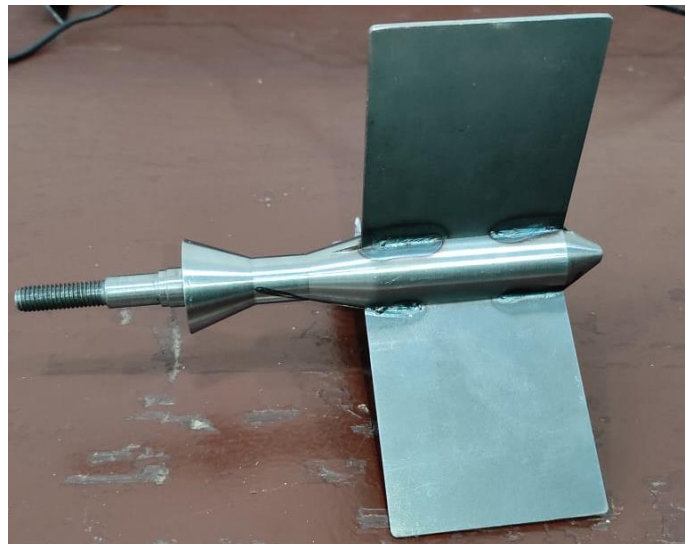
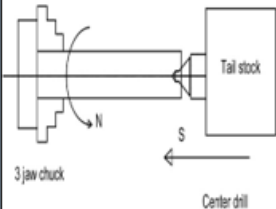
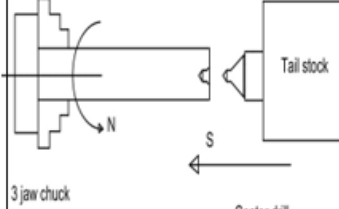
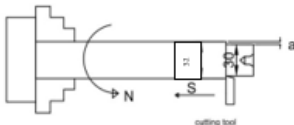
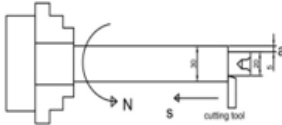
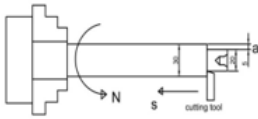
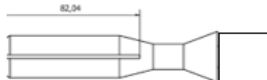
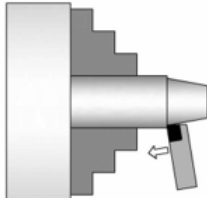
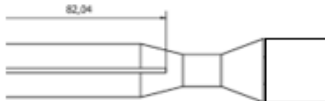
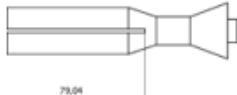


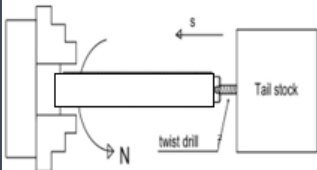
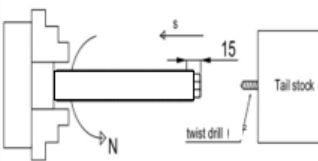
Figure 34 Upstream Assembly after Welding

Operation	Description		Cutting condition
21- centering	Before 	After 	Feed = 0.1 mm/rev -width of cut = 2.5mm -Run in = 2mm - length=5mm - N = 200 m/min. - no of strokes = 1
	Remarks -fixation: center drill is fixed in tail stock Workpiece is fixed by 3 jaw chuck -width of cut = 5mm (the length of tool tip) - objective : the center drill opens a small hole for the dead center in order to fix the W.P as it is a little bit long W.P and to continue all the following turning operations without bending concern.		$T=2+5/200^*$ 0.1 =0.14 min

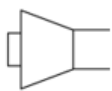
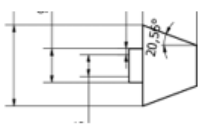
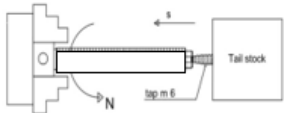
Operation	Description		Cutting condition
22- longitudi nal turning (Ø32 to Ø25)	Before 	After 	Feed = 0.2 mm/Rev -depth of cut = 1 mm -length = 145mm -Run in = 2mm - N = 400 ,500 rpm - no of strokes = ma/a = 2 T = (145+1/400*0 .3) = 1.64 min
	Remarks -fixation: 3 jaw chuck and live center -Machining allowance = $d_i - d_f / 2 = 32 - 25 / 2 = 3.5 \text{ mm}$ *depth of cut per first stroke = 2 mm /stroke *and second stroke = 0.5mm/stroke a=1 i=2		

Operation	Description		Cutting condition
23- Tapered turning	Before 	After 	Feed = 0.2 mm/Rev -depth of cut = 1 mm -length = 145mm -Run in = 2mm - N = 400 ,500 rpm - no of strokes = $\frac{ma}{a} = 2$ T = $(145 + \frac{1}{400} * 0.3) = 3.04\text{min}$
	Remarks -fixation: 3 jaw chuck and live center *depth of cut per first stroke = 2 mm /stroke *and second stroke= 0.5mm/stroke a=1 i=2 		

<u>Operation</u>	<u>Description</u>		<u>Cutting condition</u>
24- longitudi nal turning (Ø25 to Ø12.5)	Before 	After 	Feed = 0.2 mm/Rev -depth of cut = 1 mm -length = 50mm -Run in = 2mm - N = 400 ,500 rpm - no of strokes = ma/a = 4 T = 4*(8+1/400*0. 3) = 1.43min
	Remarks - fixation : 3 jaw chuck and live center - Machining allowance = $d_i - d_f / 2 = 25 - 12.5 / 2 = 6.25$ mm		

Operation	Description		Cutting condition
25- Drilling Ø7	Before 	After 	Feed = 0.1 mm/rev. - Length = 21mm. -Run in = 2mm. - N =500 rpm. - No. of strokes = 1. -T/stroke = $(9+2)*65/500*$ 0.1 = 0.76 min
	Remarks Fixation: 3-jaw chuck Depth of cut: 9mm. (width of cutter) Objectives: To open a hole in the sensor to be fixed on the M8 thread.		

<u>Operation</u>	<u>Description</u>		<u>Cutting condition</u>
26- Internal Under cutting On Ø 8	Before	After	- Feed =0.1mm/rev. - Depth of cut = 1mm -Run in = 2mm. - Run out = 2mm. - N =500 rpm. - No. of strokes = 1
	Remarks -fixation: 3 jaw chuck and live center -ma = 1mm -Depth of cut = 1 mm - objective :we make this process on the end of threaded portion of screw to provide clearance for the cutting tool during the threading.		-T = (1+2+2)/500*0.1 = =0.1 min

Operation	Description		Cutting condition
27- Internal thread Ø 17 to M18	Before 	After 	Feed = 0.1 mm/Rev -length =21mm -Run in = 2mm - N =300 - no of strokes = 1 T = (18+2/300*0.1)= 1.2 min
	Remarks -fixation: 3-jaw chuck and fixing the tool-on-tool post. -depth of cut : 0.5 mm -Note: automatically powering our feed by lead screw. **Feed is equal to the pitch of the thread=0.1mm*. -objectives: preparing for the M18 that will be fixed here. 		

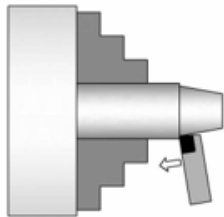
<u>Operation</u>	<u>Description</u>		<u>Cutting condition</u>
28- Tapered turning	Before 	After 	Feed = 0.2 mm/Rev -depth of cut = 1 mm -length = 145mm -Run in = 2mm - N = 400 ,500 rpm - no of strokes = $m/a = 2$ T = $(145 + 1/400 * 0.3) = 3.04 \text{ min}$
	Remarks - fixation : 3 jaw chuck * depth of cut per first stroke = 2 mm /stroke *and second stroke = 0.5mm/stroke a=1 i=2 		



Figure 35 Downstream Assembly after welding

We use Laser Welding For weld the Blades :-

We make 3 Slots on each Hanger and 5 Slots in Hub in Horizontal Milling



Figure 36 Upstream and Downstream Hanger and Hub Before Welding



Figure 39 Rotor Assembly after welding



Figure 40 Rotor Assembly after Preparation

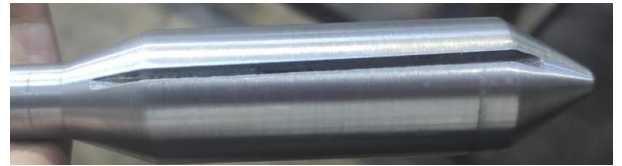


Figure 37 Slots in Hanger



Figure 38 Another View of Slots



Figure 41 Hanger with Rotor After Welding

Bolts Calculation and Selection: -

Bolt Calculations

E=210 Gpa , N=8 , Drag Force=Negligible ,mass for flowmeter =9Kg on 16 Bolt
 $\mu = 0.2$

Joint Stiffness :-

$$C = \frac{EA}{L} = \frac{E \times \frac{\pi}{4} \times (Dm^2 - a^2)}{L}$$

$$C_4 = \frac{210 \times 10^9 \times \frac{\pi}{4} \times ((\frac{47+27}{2} \times 10^{-3})^2 - (18 \times 10^{-3})^2)}{10 \times 10^{-3}} = 17.2355627 \text{ GN/m}$$

$$C_2 = \frac{210 \times 10^9 \times \frac{\pi}{4} \times ((\frac{47+33}{2} \times 10^{-3})^2 - (18 \times 10^{-3})^2)}{10 \times 10^{-3}} = 21.04552919 \text{ GN/m}$$

$$C_1 = \frac{210 \times 10^9 \times \frac{\pi}{4} \times ((\frac{33+27}{2} \times 10^{-3})^2 - (18 \times 10^{-3})^2)}{3 \times 10^{-3}} = 31.66725395 \text{ GN/m}$$

$$C_3 = \frac{210 \times 10^9 \times \frac{\pi}{4} \times ((47 \times 10^{-3})^2 - (18 \times 10^{-3})^2)}{3 \times 10^{-3}} = 62.1799726 \text{ GN/m}$$

$$\frac{1}{C_j} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \frac{1}{C_4} \rightarrow C_j = 6.527572187 \text{ GN/m}$$

Bolt Stiffness:-

$$C = \frac{EA}{L} = \frac{E \times \frac{\pi}{4} \times D^2}{L}, A_c = 1.37 \text{ cm}^2$$

$$C_1 = \frac{210 \times 10^9 \times \frac{\pi}{4} \times (16 \times 10^{-3})^2}{16 \times 10^{-3}} = 2.638937829 \text{ GN/m}$$

$$C_2 = \frac{210 \times 10^9 \times \frac{\pi}{4} \times (13.2 \times 10^{-3})^2}{12 \times 10^{-3}} = 2.39483608 \text{ GN/m}$$

$$\frac{1}{C_b} = \frac{1}{C_1} + \frac{1}{C_2} \rightarrow C_b = 1.255484183 \text{ GN/m}$$

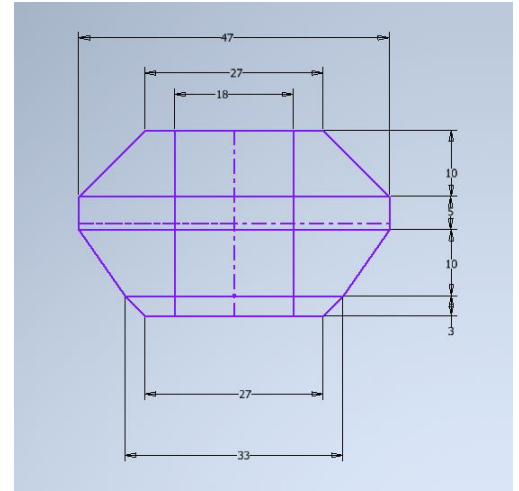


Figure 42 Influence Cone

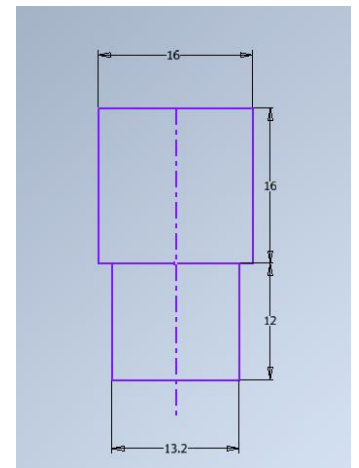


Figure 43 Bolt's dim

Forces on Bolt

$$F_{sh} = \frac{mg}{2N\mu}$$

$$F_{ex} = F_{sh}$$

$$F_{ex} = 28.125 \text{ N}$$

$$\text{Assume } F_r = 0.3 F_{ex} \rightarrow F_r = 8.4375 \text{ N}$$

$$F_{max} = F_r + F_{ex} = 36.5625 \text{ N}$$

$$F_j = \frac{C_j}{c_b + c_j} F_{ex} = 23.58816 \text{ N}$$

$$F_i = F_r + F_j = 32.1 \text{ N}$$

Bolt Selection

Working Condition

$$F_{max} = 36.5625 \text{ N}$$

$$\sigma = \frac{F_{max}}{A_c} = 267.17643 \text{ Kpa}$$

$$\tau_{max} = \sqrt{\left(\frac{\sigma}{2}\right)^2 + \tau^2} = \frac{\sigma}{2} = 133.5882 \text{ Kpa} \leq \frac{\sigma_y}{2n}, n=2$$

$$\sigma_y = 534.353 \text{ Kpa}$$

Assembly Conditions

$$T_t = F_i \frac{D_m}{2} \left(\frac{\tan \alpha + \frac{\mu}{\cos \beta}}{1 - \tan \alpha \times \frac{\mu}{\cos \beta}} \right) + F_i \frac{D_s}{2} \mu, \tan \alpha = \frac{P}{\pi D_m}, D_m = \frac{D_o + D_i}{2}, D_s = 1.3 D_o, \beta = 30, \mu = 0.2, F_i = 32.1 \text{ N}$$

$$T_t = 0.13 \text{ N.m}$$

$$\sigma = \frac{F_i}{A_c} = 234.57 \text{ Kpa}$$

$$\tau = \frac{16 T_t}{\pi D_c^2} = 287.9 \text{ Kpa}$$

$$\tau_{max} = \sqrt{\left(\frac{\sigma}{2}\right)^2 + \tau^2} = 310.8733 \text{ Kpa} \leq \frac{\sigma_y}{2n}, n=2$$

$$\sigma_y(\min) = 1.2435 \text{ Kpa}$$

So we can use 3.6 Grade and Bolt Dimension For M16× 48 × 32 × 1.5 (available)

Electrical Design

Velocity sensor :-

From the Rotor we measure (Rpm from pulse)

We use E3F-DS30C4 NPN, NO, Infrared Reflection Photoelectric Sensor

That it has fast detection response and detects nonferrous objects like Aluminium .

E3F-DS30C4 NPN Model:-

Working Principle

The E3F-DS30C4 operates based on the principle of **infrared reflection**:

1. **Infrared Emission:** The sensor emits an infrared (IR) light beam from its transmitter.
2. **Reflection Detection:** When an object enters the sensor's detection range (up to 30 cm), the IR beam reflects off the object and returns to the sensor.
3. **Signal Processing:** The sensor's receiver detects the reflected IR light and processes the signal.
4. **Output Triggering:** If the reflected signal exceeds a predefined threshold, the sensor's NPN output switches to the "ON" state, indicating the presence of an object.

Selection

The E3F-DS30C4 sensor was chosen for the following reasons:

- **High Reliability:** It provides accurate and consistent object detection, even in challenging environments.
- **High Response :** Accurate readings at high RPM
- **Easy Integration:** Compatible with most industrial control systems due to its NPN output.

General Specifications:-

Parameter	Specification
Type	Infrared Reflection Photoelectric Sensor
Output Configuration	NPN, Normally Open (NO)
Detection Range	30 cm
Operating Voltage	6–36 V DC
Maximum Current	300 mA
Response Time	1.5 ms max
Light Source	Infrared LED
Housing Material	Plastic
Protection Rating	IP67 (Dustproof and waterproof)
Operating Temperature	-30°C to +65°C

Datasheet

https://download.datasheets.com/pdfs/2013/9/4/6/24/26/674/omr_/manual/cedsax4_us_e.pdf

more info

[E3F-DS30C4 NPN, NO, Infrared Reflection Photoelectric Sensor 30Cm, 300mA, 6-36Vdc | Ampere Electronics](#)

- sensor Dimension

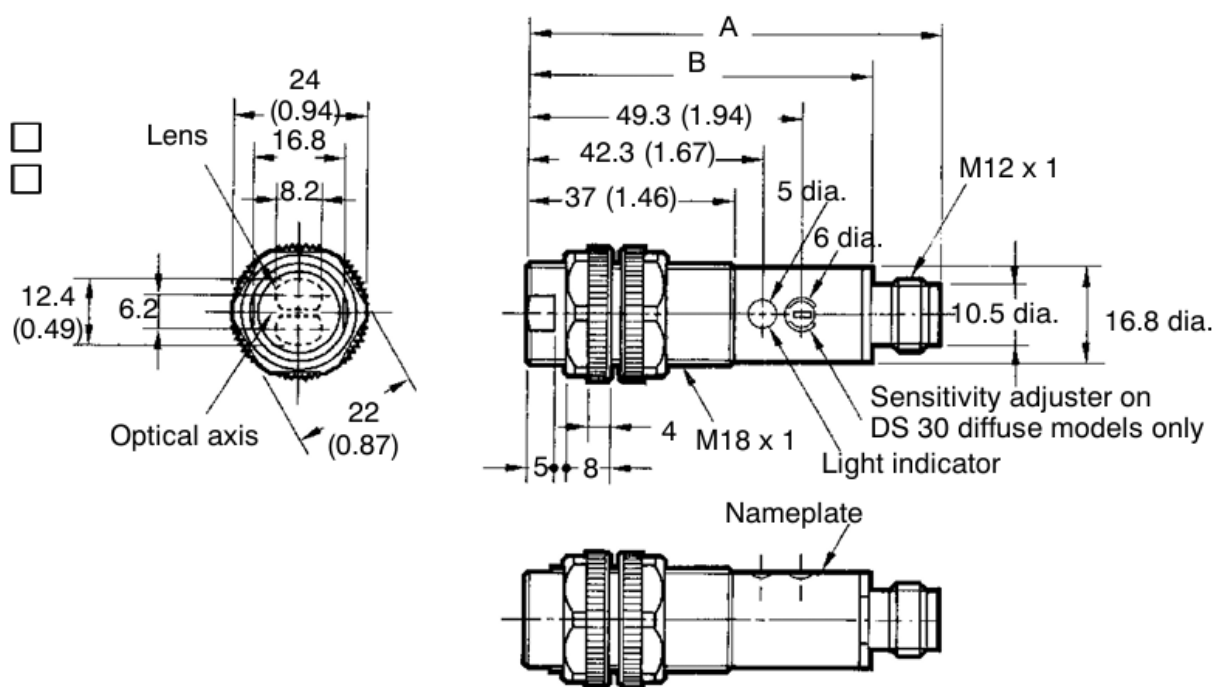


Figure 44 Sensor Dimensions

	Metal body ✕ ✕	Plastic body ✓ ✓
	E3F2-DS□□□□-□1	E3F2-DS30□4-P1
A	76 (2.99)	73 (2.87)
B	65.5 (2.58)	61.7 (2.43)

Figure 45 Figure 1 Sensor Dimensions 2



Figure 46 E3F-DS30C4 NPN

Interfacing with microcontroller (Max input =5 VDC)

- As The sensor output 0→12VDC the microcontroller needs a voltage divider circuit.
- We designed it with a Voltage regulator From 12V To 5V

- How we measure velocity from it

1. We read Pulses from the sensor that detects a rising edge (positive edge), the microcontroller starts a timer.
2. When 5 Pulses are detected, it calculates the RPM and prints it.
3. After 10 RPM calculations, it computes and prints the average RPM

Temperature & Pressure Sensor:-

We chose it because it has high sensitivity ,accuracy 0.1°C, Well calibrated using stored calibration coefficients in the E²PROM.

We can measure both Temperature and pressure with the same sensor.

Working Principle

The BMP180 is a digital pressure sensor based on piezoresistive technology. It measures both pressure and temperature using a MEMS (Micro-Electro-Mechanical Systems) sensor.

Pressure Measurement:

- The sensor contains a piezoresistive element that changes its electrical resistance when subjected to mechanical stress (pressure).
- The pressure deforms a silicon diaphragm, causing a change in resistance, which is converted into an electrical signal.
- The sensor outputs an uncompensated pressure value, which is then calibrated using stored calibration coefficients in the E²PROM.

Temperature Measurement:

- The BMP180 also includes a temperature sensor to compensate for temperature effects on pressure measurements.
- The temperature sensor measures the ambient temperature, which is used to correct the pressure readings for thermal drift.

I²C Interface:

- The BMP180 communicates with a microcontroller via the I²C interface.
- The microcontroller sends commands to start pressure or temperature measurements and reads the results from the sensor's registers.

Calibration:

- The sensor is factory-calibrated, and the calibration coefficients are stored in the E²PROM.
- These coefficients are used to compensate for offset, temperature dependence, and other non-linearities in the sensor's output.

Reason for Using the Sensor

- High Accuracy: It provides precise pressure and temperature measurements, making it suitable for applications requiring high reliability.
- Low Power Consumption: With a power consumption of only 5 µA in standard mode, it is ideal for battery-powered devices.
- Compact Size: The small LGA package allows for easy integration into compact designs.
- Wide Operating Range: It can operate in a wide temperature range (-40°C to +85°C) and measure pressure from 300 to 1100 hPa.
- I²C Interface: The I²C interface simplifies communication with microcontrollers, reducing the complexity of the system.
- Cost-Effective: It is an economical solution for high-volume applications.

How we measure P,T:-

- As the sensor is manufactured calibrated , it sends the P and T Values to the Controller with I2C Communication. P in hPa=100 Pa , Tin °C.

General Specifications

Parameter	Specification
Pressure Range	300 to 1100 hPa (equivalent to +9000m to -500m relative to sea level)
Temperature Range	-40°C to +85°C (operational), 0°C to +65°C (full accuracy)
Supply Voltage	1.8V to 3.6V (VDD), 1.62V to 3.6V (VDDIO)
Power Consumption	5 μ A at 1 sample/sec in standard mode
Resolution	Pressure: 0.01 hPa, Temperature: 0.1°C
Accuracy	Pressure: ± 0.12 hPa (typical), Temperature: $\pm 0.5^\circ\text{C}$ (typical)
Interface	I ² C (up to 3.4 MHz)
Package	LGA (3.6mm x 3.8mm x 0.93mm)
Applications	GPS navigation, weather monitoring, altitude measurement, consumer electronics



Figure 47 BMP 100 Sensor

Datasheet

<https://cdn-shop.adafruit.com/datasheets/BST-BMP180-DS000-09.pdf>

Communications

1) Remote Monitoring Station (Between Flowmeter and Mobile App)

We Use Bluetooth Module to Transfer the Data Between the Flowmeter and the mobile app

The Microcontroller send Mass flow rate and Pressure Drop



Figure 48 HC-05 Bluetooth to Serial Port Module

Datasheet

https://components101.com/sites/default/files/component_datasheet/HC-05%20Datasheet.pdf

2)Between 2 BMP, LCD and Arduino

We use I2C Expander

- The TCA9548A device has eight bidirectional translating switches that can be controlled through the I2C bus. The SCL/SDA upstream pair fans out to eight downstream pairs, or channels. Any individual SCn/SDn channel or combination of channels can be selected, determined by the contents of the programmable control register. These downstream channels can be used to resolve I2C target address conflicts. For example, if eight identical digital temperature sensors are needed in the application, one sensor can be connected at each channel: 0-7.
- The system controller can reset the TCA9548A in the event of a time-out, or other improper operation by asserting a low in the RESET input. Similarly, the power-on reset deselects all channels and initializes the I2C/SMBUS state machine. Asserting RESET causes the same reset and initialization to occur without powering down the part. This allows recovery if one of the downstream I2C buses get stuck in a low state.
- The pass gates of the switches are constructed so that the VCC pin can be used to limit the maximum high voltage, which is passed by the TCA9548A. Limiting the maximum high voltage allows the use of different bus voltages on each pair, so that 1.8V, 2.5V or 3.3V parts can communicate with 5V parts, without any additional protection. External pullup resistors pull the bus up to the desired voltage level for each channel. All I/O pins are 5V tolerant.

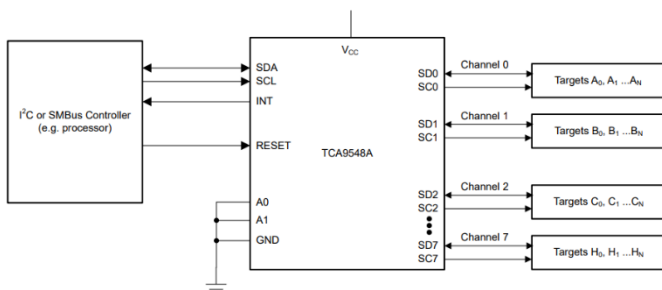


Figure 49 Simplified Application Diagram



Figure 50 TCA9548A PCA9548A I2C Expansion Module

Datasheet:-

<https://www.ti.com/lit/ds/symlink/tca9548a.pdf>

Load Profile and Battery Sizing for Arduino System

- Battery sizing is a critical process in designing an energy storage system. The procedure involves determining the size and capacity of a battery to meet specific energy needs while ensuring reliable and efficient performance. We used two-way power for the Smart Flow Meter (Power Adopter, Batteries) to ensure another power method to supply the device. Power will be supplied at rate of 12v and 1A for one Day.

Energy Demand (Power Consumption)

Components	Voltage (V)	Current (mA)	Power (W)=IV	Adjusted Power (W)
Arduino Uno	5	50	$0.05 \times 5 = 0.25$	0.28
IR Sensor	12	30	$0.03 \times 12 = 0.36$	0.36
BMP Sensor (2)	3.3	1	$0.001 \times 3.3 = 0.0033$	0.0037
16x2 LCD	5	20	$0.02 \times 5 = 0.10$	0.11
Bluetooth Module	5	50	$0.05 \times 3.3 = 0.165$	0.183
i2C Expansion Module(Multiplexer)	5	20	$0.02 \times 5 = 0.1$	0.11
Total Power Consumption = 1.0467W				

- Batteries has efficiency losses. So we included this in the calculation:
- Battery Efficiency (η): Typically 80–95%.
- Adjusted Power = Power / Battery Efficiency

Battery Capacity

Daily Energy Demand

- Daily Energy (Wh) = Total Power (W) × Hours per Day (24)
- Daily Energy (Wh) = $1.0467 \times 24 = 25.1208$ Wh/day

Battery Sizing for 1 Day

- Battery Voltage = 12V
- Battery Capacity (Ah) = Daily Energy (Wh) / (Battery Voltage × Depth of Discharge (DoD)).
- For Lithium-Ion Battery (DoD = 90%).
- Battery Capacity (Ah) = $35.61 / (12 \times 0.9) = 3.30$ Ah.

Used Battery:

- Rechargeable Li-ion Li-Polymer Battery 11.1V 12V 12.6V 3000mAh (Made in Egypt) 12V 3000mAh 18650 Li-ion lithium Rechargeable Battery Pack

Technical Parameters

Nominal voltage: 12v

Input voltage: 12.6V

Output voltage: 8V to 12.6V

Battery capacity: 3000mah (3Ah), deviation $\pm 5\%$,

Size: 70*56*35MM, deviation ± 5 mm,

Packing: Blue PVC

With protection circuit: have overcharge, over discharge, over current, short circuit protection functions the battery requirements

1: need li-ion battery chargers for; Constant current constant voltage;

2: the charging voltage: 12.6V

3: charging current: 1A

4: supporting the charger is charging when the red light is full of green light

Battery Discharge Requirements:-

1: input: 5.5 x 2.1 plug wire

2: output: 2 line, about 115 mm long

3: the maximum instantaneous current: 3A

4: maximum working current: 3A

5: output input if there is any other requirement can be contact with the owner, the above parameters for the default delivery applicable scope

All the 12 v electric equipment, washing equipment, coal, hernia lamp, instead of the 12 v battery, instruments and meters, amplifiers, monitoring, electric class, all sorts of article 12 v lamp.

lithium battery advantage

1. Small volume, light weight, large capacity, environmental protection, no memory effect.

2. Is the power supply for portable electronic devices, but more delicate, lithium-ion batteries in use, please note the following points.

Lithium battery common sense

1. You must use a 12.6 V li-ion battery charger charging standard.

2. The ban on large current discharge.

3. Prohibited discharge, single battery voltage should not be below 2.5 V.

4. It is strictly prohibited to disassemble, short circuit batteries, most likely caused by short circuit on fire.

5. Use temperature 0 to 60



Figure 51 -Rechargeable Li-ion Li-Polymer Battery

Electrical Schematic

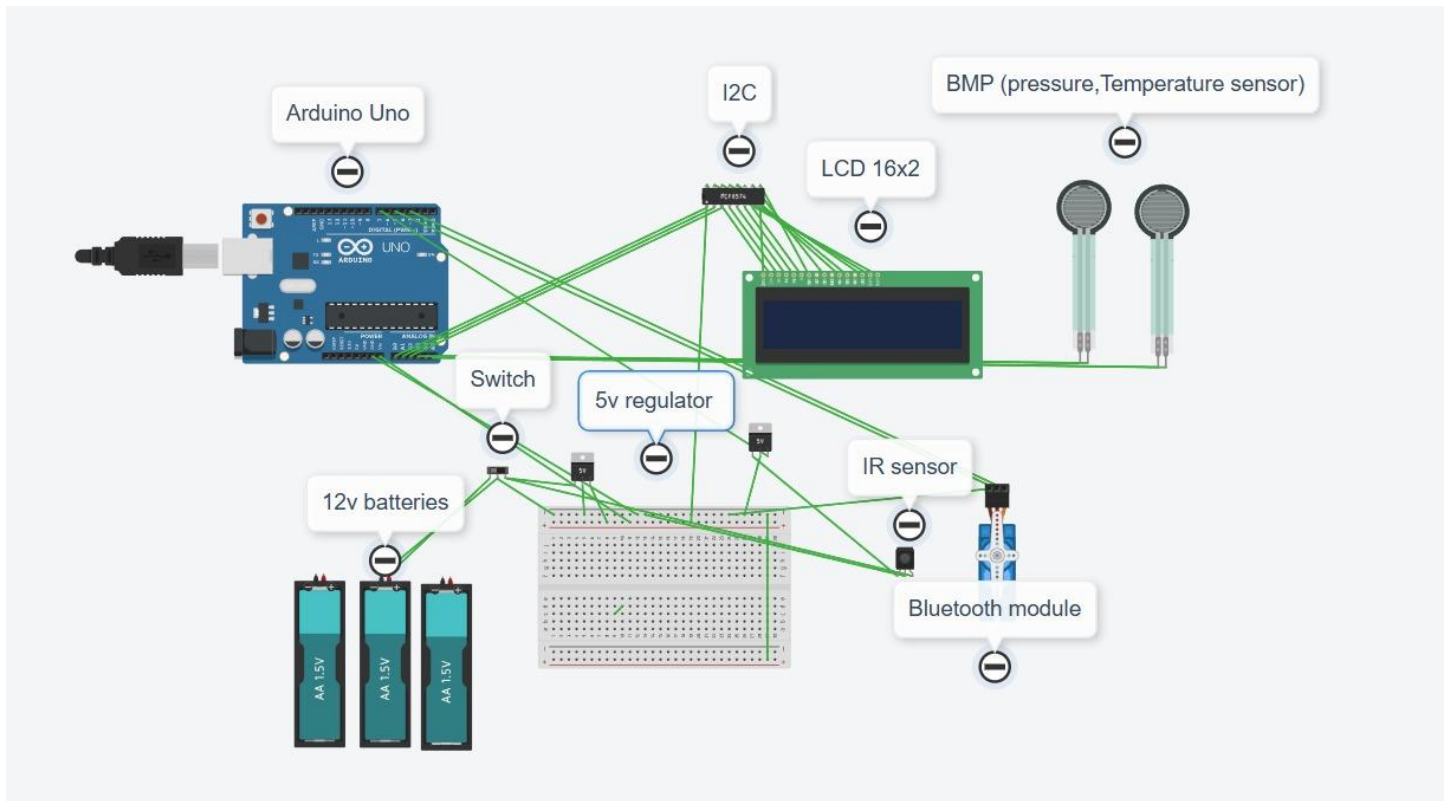


Figure 52 Electrical Schematic

Calibration Of Smart Flow Meter

Calibration Setup



Figure 53 Setup 1



Figure 54 Setup 2

Results:-

1)Rpm, Velocity of the flow and Mass Flow Rate

Rpm	Vin	M°
123	13.3	0.017392
179	15	0.019615
258	19	0.024846
323	23	0.030077
394	26	0.034
450	29	0.037923

Using polynomial regression :

We used a second Degree best Fit curve to describe the behaviour of the system.

$$Y = -0.000000106x^2 + 0.000129x$$

Y represents mass flow rate

X represents the RPM

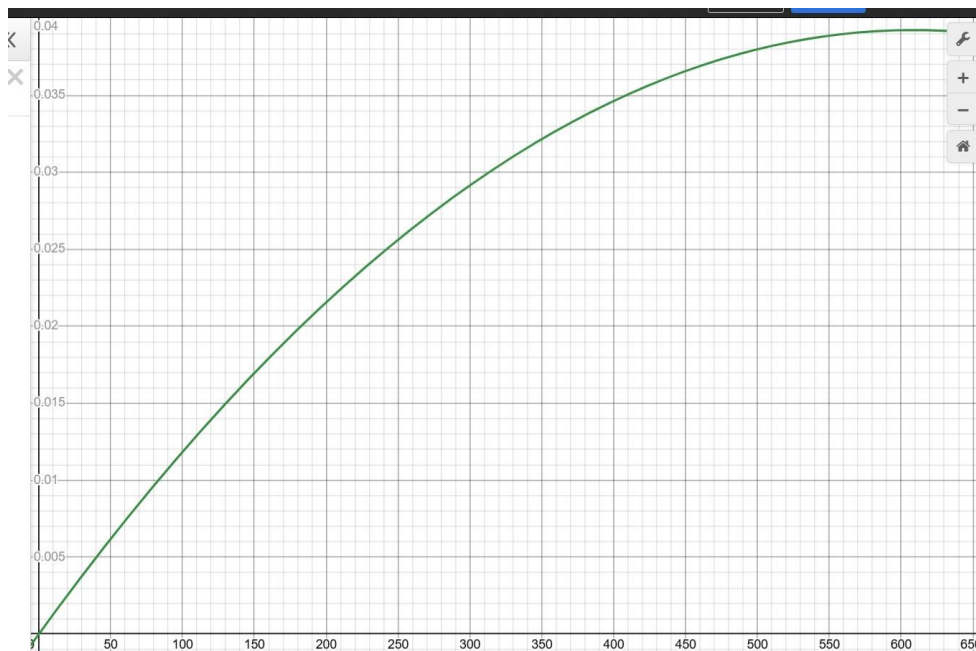


Figure 55 graph of the Function



The brochure features a dark blue background with a large blue diamond shape on the left side. Inside the diamond, the text 'Price Starts at 30,000EGP' is displayed in white, with '30,000EGP' in a larger font. Below this, a yellow button with the text 'Book Now' is visible. In the top left corner of the brochure, there is a circular logo for 'ASU Team' with a stylized building and the year '1958'. On the right side, there are two images: a 3D rendering of the Smart Flow Meter and a photograph of the physical device. The physical device is a stainless steel pipe with a black electronic control box mounted on top. The control box has a small LCD screen and a label that reads 'Smart Flow Meter' and 'ASU Team'. Below the images, the text 'The Perfect Device to measure mass flow rate' is followed by a description: 'Simple smart device measures three parameters for the flow and can accurately calculate and monitor mass flow rate in real time response.' At the bottom left, the text 'Call For Booking :' is followed by a phone icon and the number '01004193456'. At the bottom right, the website 'www.asuteam1mc.com' is listed.

 Price Starts at
30,000EGP
Book Now

SMART FLOW METER

The Perfect Device to measure mass flow rate
Simple smart device measures three parameters for the flow and can accurately calculate and monitor mass flow rate in real time response.

Call For Booking :
 **01004193456**

www.asuteam1mc.com

Figure 56 Brochure of our Project

Problems

Low Inertia Requirement:

The application necessitates a rotor with minimal rotational inertia.

- **Material Selection:** Aluminum was chosen as the primary material due to its low density, but this introduces challenges.
- **Blade Thinness:** To further minimize inertia, the rotor blades need to be thin. However, this makes them difficult to weld using conventional methods.

Welding Challenges:

- High heat input from welding can cause significant distortion and warping of the thin blades.
- Risk of material weakening due to excessive heat.
- Difficulty in achieving consistent penetration depth in thin materials.

Solution:

Laser Welding:

Employ laser welding to address the welding challenges:

- **Precision:** Laser welding offers high precision, minimizing heat-affected zones and reducing distortion.
- **Controllability:** Precise control of laser parameters allows for localized heating and minimizes the risk of material damage.

Recommendations

1) Laser Welding for Minimal Flow Disturbance:

- Traditional welding methods can introduce distortions and protrusions into the flow path, affecting the accuracy and consistency of flow measurements.
- Laser welding offers a precise and localized heat input, minimizing the heat-affected zone and reducing the risk of distortion. This ensures a smooth, uninterrupted flow profile within the flowmeter, maintaining accuracy.

2) Tachometer Sensor(magnetic or optical) for Accurate Flow Measurement:

- IR sensors can be susceptible to ambient light interference and may not provide the same level of precision in measuring rotational speed as a tachometer.
- A tachometer directly measures the rotational speed of the Flowmeter. This provides a more accurate and reliable reading, directly correlating to the flow rate.

3) XGZP190 Pressure Sensor for Precise Pressure Measurement:

- For the required accuracy or stability: The XGZP190 is a high-accuracy, piezoresistive pressure sensor known for its stability and reliability. It offers a precise measurement of the pressure differential across the Flowmeter, which can be valuable data for flow rate calculation and system diagnostics.

[datasheet](#)

[https://cfsensor.com/wp-content/uploads/2022/11/XGZP190- Pressure-Sensor-V2.3.pdf](https://cfsensor.com/wp-content/uploads/2022/11/XGZP190-Pressure-Sensor-V2.3.pdf)

4) Dedicated Power Supply with Battery Backup:

- Using an adapter for the primary power source provides a stable supply.
- Adding a battery backup ensures continuous operation even during power outages.

5) NRF Module for Wireless Communication:

For wireless data transmission.

- The NRF module allows for wireless communication between the flowmeter and a remote monitoring station or data logger.

6) Raspberry Pi or PLC for Advanced Control and Data Processing:

- Arduino Uno might be limited in processing power, memory, and communication capabilities for this application.
- A Raspberry Pi (for more flexible software and data processing) or a PLC (for robust industrial control) provides more advanced capabilities:
 - o Data Logging: Store flow rate, pressure, Temperature and other data over time.
 - o Control Algorithms: Implement more complex flow control algorithms or integrate with other systems.
 - o Communication: Handle wireless communication with the NRF module and potentially other interfaces.
 - o Visualization: Generate real-time displays of flow data (especially with a Raspberry Pi).

Conclusion

We Designed A Smart Flowmeter With high precision

And has nonlinearity relation

By the end of the project, We gained experience in:

- Interdisciplinary Collaboration: Integrate mechanical and electrical systems.
- Problem-Solving Skills: Address design challenges and optimizing systems.
- Hands-On Experience: Build a functional prototype from concept to testing.
- Presentation Skills: Communicate technical information effectively.

Cost Analysis

Mechanical

Machined Parts (Raw Material)

No	Part	Cost
1	Pipe	300
2	Steel 37 Rod (D32*700mm)	300
3	Aluminium 6061 Rod (D33*100mm)	150
4	Aluminium sheet (150*80mm)	100
5	Steel sheet (400*80mm)	200

Standard Parts

No	Part	Cost
1	Snap ring 160	100
3	Snap ring 19	20
4	2 Gasket	0
5	2 Flanges	400
6	2 Deep groove ball Bearing 6800	140

Electrical

No	Part	Cost
1	2*BMP180	240
2	E3F-DS30C4 IR NPN	350
3	ATC9548A I2C Module	130
4	Rechargeable Li-ion Li-Polymer Battery	500

Total Cost = 2950 LE

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